# THE ROLE OF CRISIS PREGNANCY CENTERS IN FERTILITY DECISIONS

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**Abstract**

Crisis pregnancy centers (CPCs) are the most widespread anti-abortion intervention in the United States, yet no published causal evidence exists on whether they affect fertility outcomes. Using a 30-year county panel from North and South Carolina and a simulated instruments strategy that addresses endogenous CPC placement, I estimate that one additional CPC per 10,000 women reduces abortion rates by 13–14 percent. The result is robust across functional forms, weighting schemes, and alternative estimators. Pregnancy rates do not respond to CPC presence, indicating that CPCs redirect pregnancy outcomes rather than prevent pregnancies. A marital decomposition shows that the substituting births concentrate among married women aged 25–34, while the abortion reductions are pervasive across unmarried age groups. IV estimates substantially exceed OLS, consistent with positive selection into CPC placement.

**Keywords:** Fertility, abortion, birth, crisis pregnancy centers, simulated instruments

JEL Classification: J12, J13, C15, C36, and K23

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# 1 Introduction

Crisis pregnancy centers (CPCs) are the most widespread anti-abortion intervention in the United States. More than 2,500 centers operate nationwide, outnumbering abortion clinics roughly three to one (Jones & Jerman, 2017). CPCs counsel pregnant women toward carrying to term, offering pregnancy testing, material support, and community networks (Care Net, 2018). They receive approximately $89 million in public funding annually (Kruesi, 2022),[2] and by their own account prevent tens of thousands of abortions each year.[3] Yet no published causal evidence exists on whether they actually affect fertility outcomes. The existing literature focuses on supply-side restrictions: clinic closures raise driving distances and reduce abortions by roughly 20 percent (Lindo, Myers, Schlosser, & Cunningham, 2020), travel distance substantially alters reproductive control (C. K. Myers, 2017), and reduced access to both abortion and family planning services jointly affects fertility (Fischer, Royer, & White, 2018). These papers share a common structure: an exogenous shock to the cost of obtaining an abortion. The demand side, where institutions try to change women's choices rather than their choice sets, has received less attention.

This paper asks whether CPC presence causally reduces abortion rates. I use a 30-year county panel from North and South Carolina (1990–2019) and a simulated instruments strategy that addresses endogenous CPC placement. A discrete-time hazard model of CPC entry, estimated from predetermined county characteristics, generates predicted opening probabilities. Forward-simulating 1,000 counterfactual expansion paths and averaging yields the instrument, which converges to expected CPC presence conditional on observable county history. The identifying assumption is predetermination: conditional on rich historical observables and two-way fixed effects, the residual timing of CPC openings is orthogonal to contemporaneous fertility shocks. The exclusion restriction follows from the law of iterated expectations, without requiring a separate argument about the instrument's channels of influence (§4.3). Monte Carlo validation confirms the estimator recovers the true causal parameter under severe designed endogeneity, even when the hazard model

[2] State appropriations to CPCs accelerated sharply after *Dobbs*, totaling nearly $489 million in the two years following the decision (CBS News, 2024). CPC counts have grown 3.3 percent nationally since *Dobbs*, with expansion concentrated in states that banned abortion and states with dedicated CPC grant programs (Lambert, Luisi, Johnson, Baugh, & Swartzendruber, 2025).

[3] According to self-reported data from Care Net, their 1,100 affiliated CPCs prevented 677,248 abortions between 2008 and 2017, or about 1 in 10 abortions (Care Net, 2022).

is deliberately misspecified.

One additional CPC per 10,000 women reduces abortion rates by 13–14 percent in two-stage least squares (2SLS) estimation. The baseline estimate is $-0.144$ ($F = 170$). Absorbing trending confounders via $X_c\times$ year FE interactions yields $-0.130$ ($F = 164$).[4] The result is robust across functional forms, weighting schemes, control sets, and alternative estimators including Poisson control function and augmented synthetic control. Instrumental variable (IV) estimates roughly double estimates from ordinary least squares(OLS), consistent with positive selection into CPC placement.

The pregnancy rate evidence narrows the mechanism. The pregnancy rate—abortions plus births plus fetal deaths—does not respond to CPC presence at any age group or in any specification, ruling out deterrence (upstream behavioral change that prevents pregnancies) as the primary channel. Combined with the robust abortion reductions and positive (if imprecise) birth effects, the accounting is consistent with substitution: CPCs change how pregnancies are resolved, not whether they occur. An exploratory marital decomposition (Appendix F.4) reveals that abortion reductions are pervasive among unmarried women across all adult ages, while the corresponding birth increases concentrate among married women aged 25–34.

The paper makes two contributions. First, it provides the first causal estimates of CPC effects on fertility, placing demand-side interventions on equal quantitative footing with the supply-side literature. Lindo et al. (2020) estimate that clinic closures reduced abortions by roughly 20 percent; the CPC estimates are of comparable magnitude. Second, the pregnancy accounting narrows the mechanism: CPCs do not reduce pregnancies, so the averted abortions become births. Because CPCs change how pregnancies are resolved rather than whether they occur, the relevant welfare question concerns the marginal woman redirected from abortion to birth, not averted pregnancies (§6). States have substantially increased CPC funding since *Dobbs* (Lambert et al., 2025), and the mechanism evidence implies that this funding generates additional births whose welfare consequences depend on wantedness. The simulated instruments methodology extends Currie and Gruber (1996) to a dynamic, path-dependent setting, connecting to optimal instruments theory

[4]Controlling for twice-lagged abortion rates attenuates the coefficient to $-0.098$, providing a lower bound under partial departures from predetermination. This specification addresses the concern that past fertility shocks predict both CPC entry and current outcomes, at the cost of absorbing some of the causal variation.

(Chamberlain, 1987; Newey, 1990) and the formula instruments framework of Borusyak, Hull, and Jaravel (2025).

# 2 Background

## 2.1 What CPCs Do

CPCs offer pregnancy testing, counseling from a pro-life perspective, and material aid such as baby clothes, cribs, diapers, and direct financial assistance (Care Net, 2018). When a pregnancy is confirmed, many CPCs provide limited ultrasounds intended to inform about gestational age and viability. CPCs also offer adoption referrals and abstinence-focused sex education, particularly to teenagers.[5] The vast majority are faith-based organizations affiliated with evangelical Christian or Catholic networks such as Care Net, Heartbeat International, or the National Institute of Family and Life Advocates (Swartzendruber, Mercaldo, Newton-Levinson, & Rathbun, 2018). Most are not licensed medical facilities, meaning medical ethics rules and patient privacy laws do not apply.

The existing empirical literature on CPCs is entirely descriptive. Rice, Chakraborty, Keder, Turner, and Gallo (2021) find that 13.5 percent of surveyed women in Ohio had visited a CPC, with attendance higher among Black women, lower-income women, and women without college degrees. Cartwright, Tumlinson, and Upadhyay (2021) find that 13.1 percent of women searching for abortion services online visited a CPC during their pregnancy, and that CPC visitors were 21 percentage points less likely to have had an abortion. That 21-point gap is an association confounded by selection: women who visit CPCs may already be ambivalent about abortion, or CPCs might locate where abortion demand is high for unobserved reasons. Both channels bias OLS, motivating the IV approach in §4.2. Swartzendruber, Solsman, and Lambert (2021) find that 58.5 percent of emerging adults who visited a CPC in Georgia did not know that the center opposes abortion, suggesting that CPC effects may operate through the counseling environment, ultrasound provision, or material support rather than through explicit anti-abortion messaging.

[5] Abstinence education is one channel through which CPCs could affect upstream behavior and reduce pregnancies (the deterrence channel in §2.2). The weight of evidence suggests this channel is weak. Kohler, Manhart, and Lafferty (2008) find that abstinence-only education does not reduce the likelihood of intercourse. Trenholm et al. (2008) report no differences in teen sexual activity or unprotected sex. Carr and Packham (2017) find that state-level abstinence mandates have no effect on teen birth or abortion rates. Cannonier (2012) finds a reduction in birth rates only among white 15–17-year-olds. Whether CPC-delivered abstinence counseling differs in effect from state-mandated curricula is an open question, but the pregnancy null reported in §5.2 is consistent with this prior.

These channels all operate conditional on pregnancy, consistent with the substitution mechanism developed in §2.2.

Most research on reproductive access examines supply-side restrictions—clinic closures, mandatory waiting periods, travel distance—that directly affect women who are already pregnant (Fischer et al., 2018; Lindo et al., 2020; C. K. Myers, 2017; **?**). A demand-side intervention like a CPC shapes the decision to become pregnant in the first place, blurring the line between substitution and deterrence. Whether CPCs primarily redirect existing pregnancies toward birth or also reduce pregnancy initiation is ultimately an empirical question, and one that the data in this paper can distinguish.

## 2.2 Theoretical Framework and Predictions

CPCs provide counseling, material support, and community networks to pregnant women considering abortion. This institutional structure suggests two polar mechanisms through which CPCs may affect fertility outcomes, distinguishable by their effect on pregnancy rates.

The first is *substitution*. CPC counseling persuades women who would otherwise have terminated to carry to term. This channel implies a one-for-one substitution between abortions and births conditional on pregnancy. The pregnancy rate—the sum of abortions, births, and fetal deaths—is unchanged, because the pregnancy occurs regardless of how it is resolved.

The second is *deterrence*. If CPCs change upstream behavior (contraception, sexual activity, or relationship decisions), women who would have become pregnant do not, reducing both abortions and pregnancies. Under pure deterrence, abortions fall, births do not rise, and pregnancy rates decline.

These two mechanisms generate an empirical test. Under substitution, $\Delta\text{pregnancies} \approx 0$ and $\Delta\text{births} \approx -\Delta\text{abortions}$. Under deterrence, $\Delta\text{pregnancies} < 0$ and $\Delta\text{births} \approx 0$. The distinction also matters for welfare: under substitution, the relevant question is whether the marginal woman redirected from abortion to birth is better or worse off; under deterrence, the question is about averted pregnancies. The mechanism evidence therefore determines which welfare framework applies (§6).

In practice, both mechanisms may operate simultaneously, and the relative importance may vary by age and marital status. A third possibility—community spillovers that increase births

among married women who are not CPC clients—could produce birth increases without corresponding abortion reductions in the same group. An exploratory marital decomposition (Appendix F.4) asks where the substituting births land across demographic groups. The framework organizes the heterogeneity analysis, but the county-level data cannot pin down which behavioral channel dominates (§6.3).

### 2.3 The CPC Network in North and South Carolina

The CPC network has expanded steadily since abortion legalization in 1973. In my sample, the number of unique CPC locations in North and South Carolina grew from roughly 60 in 1990 to 146 by 2019.[6] Growth was gradual rather than concentrated around particular policy events, reflecting decentralized decisions by individual organizations responding to local conditions. By 2019, the typical county resident lived within 8 miles of a CPC, down from over 20 miles in 1990. CPCs and abortion providers overlap geographically: by 1990, all but one county with an abortion provider already had a nearby CPC, and subsequent CPC expansion targeted underserved areas farther from existing centers (§A). The absence of discrete policy shocks driving CPC expansion is what makes standard difference-in-differences designs inadequate for this setting and motivates the simulated instruments approach developed in §4.2.

### 2.4 Policy Environment

Throughout the study period (1990–2019), abortion was constitutionally protected and relatively accessible in both states. North Carolina required parental consent for minors beginning in 1995 and enacted a 72-hour waiting period in 2015. South Carolina required parental consent beginning in 1990 and introduced a 24-hour waiting period in 1995. Neither state enacted clinic-level TRAP regulations severe enough to close facilities during the sample period, and the number of abortion providers was relatively stable (20 in 1990, 23 in 2019). The FDA approval of mifepristone in 2000 introduced medical abortion as an alternative to surgical procedures, lowering the cost of obtaining an abortion in the second half of the sample period. I control for state-level parental

[6] Table I reports population-weighted county-level means of 1.12 CPCs in 1990 and 2.79 in 2019, which are higher than the unweighted means (0.40 and 0.97) because larger counties host more centers. The treatment variable counts all CPCs within a county's catchment area, including centers near county borders that serve multiple counties, so the sum of county-level counts exceeds the number of unique locations statewide.

involvement and targeted restrictions on abortion providers (TRAP) policy changes directly, and the CPC coefficient is unchanged (Table A20). The stability of the provider landscape means that the CPC effects estimated here reflect demand-side persuasion in a setting where women retained meaningful access to abortion services, not supply-side restrictions operating simultaneously.

# 3 Data

I focus on North and South Carolina because these states provide fertility counts by age, ethnicity, and county of residence beginning in 1990, allowing me to study CPC openings and fertility outcomes over a 30-year horizon (1990–2019). The analysis sample covers 146 counties observed annually, yielding 4,380 county-year observations. Summary statistics are reported in Table I.

## 3.1 CPC and Abortion Provider Data

I construct a longitudinal registry of CPC and abortion provider locations in NC and SC, including precise addresses and dates of operation. The dataset contains 288 CPC locations associated with 138 CPC organizations and 43 abortion provider locations.[7] I validate the registry by cross-referencing Internal Revenue Service (IRS) filings, organizational databases, and provider directories. Abortion provider data combine Title X grant recipient records, state licensing records, and provider lists from the National Abortion Federation and Planned Parenthood, verified against Myers' Abortion Facility Database (C. Myers & Ladd, 2020).

The main treatment variable is the lagged number of CPCs per 10,000 women ages 10–44 in a county.[8] Over the sample period, mean CPC presence roughly doubled (from 1.1 to 2.8 per county) while mean distance to the nearest CPC fell from 20 miles to 8 miles (Table I).

[7] The number of locations exceeds the number of CPC organizations because some organizations operate multiple facilities and because address changes are tracked.

[8] A typical county in the sample has approximately 10,000 women of childbearing age. Adjusting by the population in the service area follows Lindo et al. (2020). The treatment is lagged one year to avoid simultaneity and to reflect the institutional reality that CPC effects operate through established community presence rather than contemporaneous opening.

### 3.2 Fertility Outcomes

The primary outcomes are the log abortion rate and log birth rate per 1,000 women in a county and year. Counts are obtained from administrative vital records provided by the NC and SC vital statistics offices, reported by county of residence, age group (10–19, 20–24, 25–29, 30–34, 35–44), and ethnicity (white/nonwhite). I also construct a pregnancy rate per 1,000 women, defined as the sum of abortion counts, birth counts, and fetal death counts in each county-year cell.[9] This variable provides a direct test of the mechanism: under substitution, pregnancy rates are unchanged by CPC presence; under deterrence, they decline (§2.2).

Two features of the data merit note. NC resident abortion data include abortions performed in any state, while SC data are limited to abortions performed in SC, NC, or Georgia, attenuating estimated CPC effects in South Carolina.[10] County-level abortion rates declined from 16.8 per 1,000 women in 1990 to 8.6 in 2019; teen rates fell from 19.0 to 2.9.

For the marital decomposition, I construct county-age-year married and unmarried female population estimates by interpolating marriage rates between Census/ACS benchmark years (1990 STF3, 2000 SF3, and ACS 5-year estimates for 2009, 2013, 2018, and 2019) and applying the interpolated shares to NBER intercensal population counts. These marital-specific populations serve as denominators for the Poisson exposure variable, ensuring that marital-specific rate estimates reflect changes per married or unmarried woman rather than per woman overall.

The empirical specifications control for time-varying county characteristics that enter both the hazard model (§4.2.1) and the second-stage equation, including distance to the nearest abortion provider, demographics, economic conditions, political ideology, and religious composition (Table I lists the full set). All controls are lagged one year. County and year fixed effects are included in all specifications.

[9] Fetal deaths are a small share of pregnancies (roughly 1 percent of the sum) and do not meaningfully affect the results. Miscarriages are not fully reported and are excluded, so the pregnancy rate is an undercount of total conceptions.

[10] At the county-year level, fewer than 1 percent of abortion observations are zero; these are dropped from the log specification. The Poisson control function retains all observations. Zero-count shares rise substantially in demographic subgroup cells (up to 39 percent for marital-age cells), motivating the Poisson estimator for subgroup decompositions (Appendix F.3, Appendix F.4).

Table I: Summary Statistics

| | 1990–2019 | | 1990 | | 2019 | |
|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD |
| ***Abortion rate (per 1,000 women)*** | | | | | | |
| Total | 11.07 | (4.90) | 16.81 | (5.95) | 8.63 | (3.14) |
| Age 10–19 | 7.86 | (5.35) | 19.02 | (6.56) | 2.91 | (0.99) |
| ***Birth rate (per 1,000 women)*** | | | | | | |
| Total | 45.87 | (15.03) | 49.45 | (13.39) | 42.02 | (13.44) |
| Age 10–19 | 22.06 | (10.28) | 34.09 | (8.86) | 9.73 | (3.63) |
| ***Pregnancy rate (per 1,000 women)***[b] | | | | | | |
| Total | 57.0 | (17.9) | 66.9 | (16.5) | 51.1 | (15.1) |
| ***Crisis Pregnancy Centers*** | | | | | | |
| No. of CPCs | 2.15 | (2.46) | 1.12 | (1.39) | 2.79 | (2.80) |
| CPCs per 10,000 women | 0.35 | (0.39) | 0.21 | (0.30) | 0.40 | (0.42) |
| Distance to nearest CPC (miles) | 11.61 | (14.30) | 20.33 | (20.84) | 7.78 | (9.52) |
| Annual CPC revenue ($000s)[a] | 463 | ( 1,013) | 302 | ( 711) | 797 | ( 1,313) |
| ***County Characteristics*** | | | | | | |
| Female population age 10–44 | 67,406 | (69,966) | 48,246 | (42,286) | 88,599 | (92,885) |
| Non-white share | 0.32 | (0.15) | 0.29 | (0.15) | 0.34 | (0.15) |
| Share age 10–19 | 0.27 | (0.03) | 0.26 | (0.03) | 0.28 | (0.02) |
| Nearest abortion clinic (miles) | 27.33 | (25.90) | 36.60 | (28.31) | 30.73 | (26.33) |
| Unemployment rate | 5.95 | (2.66) | 4.36 | (1.54) | 3.56 | (0.79) |
| U.S. House GOP vote share | 0.53 | (0.20) | 0.45 | (0.24) | 0.52 | (0.18) |
| Protestant share | 0.54 | (0.12) | 0.60 | (0.12) | 0.51 | (0.11) |
| Catholic share | 0.04 | (0.03) | 0.02 | (0.02) | 0.04 | (0.03) |
| No. of county-year obs. | 4,380 | | 146 | | 146 | |

*Notes:* Population-weighted county-level means; standard deviations in parentheses. 146 counties observed annually, 1990–2019.

*Sources:* Abortion counts from NC and SC vital statistics offices (county of residence). Birth and fetal death counts from NBER Vital Statistics Natality and Fetal Death files. CPC locations from author-constructed registry cross-referenced with IRS filings, Care Net and Heartbeat International directories, and state licensing records. CPC revenue from IRS Form 990 filings via NCCS Core Files; [a]The 1990 column for CPC revenue reports 1997 values (earliest year with non-trivial IRS filing data for matched CPCs). [b]Pregnancy rate = abortion rate + birth rate + fetal death rate. Abortion provider locations from Title X grant records, state licensing records, National Abortion Federation and Planned Parenthood directories, and Myers' Abortion Facility Database (C. Myers & Ladd, 2020). Population and demographics from NBER intercensal population estimates (SEER). Unemployment rate from Bureau of Labor Statistics, Local Area Unemployment Statistics. U.S. House GOP vote share from CQ Voting and Elections Collection. Religious adherence from Association of Religion Data Archives, Religious Congregations and Membership Study (1990, 2000, 2010), linearly interpolated.

# 4 Empirical Framework

The research question is whether CPC presence causally affects fertility outcomes. I estimate the following structural equation separately for abortion rates, birth rates, and pregnancy rates:

$$y_{c,t} = \theta_0 + \theta_1 \text{CPC}_{c,t} + \theta_2' X_{c,t} + \mu_c + \lambda_t + u_{c,t}, \tag{1}$$

where $y_{c,t}$ is the natural log of the outcome (abortion rate, birth rate, or pregnancy rate) in county $c$ and year $t$; $\text{CPC}_{c,t}$ measures CPC exposure; $X_{c,t}$ is a vector of time-varying county characteristics; and $\mu_c$ and $\lambda_t$ denote county and year fixed effects, respectively. Estimating $\theta_1$ by OLS requires strict exogeneity: $E[u_{c,t} \mid \{\text{CPC}_{c\tau}, X_{c\tau}\}_{\tau=1}^{T}] = 0$. This condition fails under strategic placement.

## 4.1 The Endogeneity Problem

Strategic CPC placement creates two endogeneity problems. First, CPCs may target counties based on unobserved confounders correlated with fertility outcomes, so that OLS conflates the causal effect with selection bias. Second, treatment assignment is a dynamic, path-dependent process: the probability a county receives its $k$th CPC depends on the entire history of past placements, compounding over 30 years into a nonlinear function of past observables and organizational shocks. No discrete policy shocks or regulatory discontinuities are available for identification.

## 4.2 Instrument Construction

### 4.2.1 Hazard Model of CPC Entry

I model CPC opening probabilities using a discrete-time logit:

$$p_{c,t} = P(\text{NewCPC}_{c,t} = 1 \mid W_{c,t-1}) = \frac{\exp(\beta' W_{c,t-1})}{1 + \exp(\beta' W_{c,t-1})} \tag{2}$$

Counties can receive multiple CPCs; the model allows repeated events by conditioning on the accumulated stock $\text{CPC}_{c,t-1}$.[11] The conditioning set $W_{c,t-1}$ includes lagged CPC stock, distance to nearest CPC, lagged abortion rates, abortion provider presence, demographics, religious compo-

[11] This extends the single-spell hazard to allow multiple openings following Mroz (2012). Details are in Appendix **??**.

sition, political ideology, unemployment, county fixed effects, and a sixth-degree polynomial time trend. Full specification details and hazard model estimates appear in Appendix **??**.

#### 4.2.2 Forward Simulation

I construct the instrument by forward-simulating counterfactual CPC expansion paths. Starting from observed 1990 conditions, I estimate the discrete-time hazard model of CPC openings based on predetermined county characteristics (§4.2.1), then simulate 1,000 counterfactual expansion paths. Each simulation draws independent shocks and uses predicted opening probabilities to determine CPC entry, updating county characteristics recursively each year. Averaging simulated CPC presence across draws yields the conditional expectation of CPC presence given predetermined characteristics:

$$\mathrm{IV}_{c,t} = \frac{1}{S}\sum_{s=1}^{S} \mathrm{CPC}_{c,t}^{(s)} \xrightarrow{p} \mathbb{E}[\mathrm{CPC}_{c,t} \mid W_{c,t-1}] \tag{3}$$

where $W_{c,t-1}$ denotes the full set of lagged county characteristics entering the hazard model. Path dependence makes closed-form computation intractable: CPC presence at time $t$ depends on the full history through $t-1$, and Monte Carlo integration via forward simulation provides a consistent approximation (McFadden, 1989).

The simulation is a computational device: the simulated shocks are computer-generated random numbers, and averaging 1,000 draws eliminates them by the law of large numbers, leaving a deterministic function of $W_{c,t-1}$. No actual organizational shocks enter the formula. As a diagnostic, regressing $\mathrm{IV}_{c,t}$ on predetermined observables (Table A9) shows that county and year fixed effects capture 77 percent of the instrument's variation (Table A9, Column 2); adding the linear hazard covariates raises the $R^2$ to 0.80, and further adding quadratic terms and interactions raises it to 0.82. The remaining variation reflects simulation noise and higher-order nonlinear structure from the logit compounding recursively over 30 years. Neither component threatens the exclusion restriction: simulation noise is mechanically independent of structural errors, and the uncaptured nonlinearity is still a function of $W_{c,t-1}$. The identifying assumption is economic: predetermined

observables do not forecast current fertility shocks (§4.3).[12]

The hazard model is what makes the instrument a strong predictor after partialling out linear controls. If a CPC opened in 2003, that changes the probability of another opening in 2010, because it changes the local CPC stock, the distance to the nearest center, and the remaining unserved population. The logit link, cumulative stock, recursive distance updating, and compounding across 30 years generate cross-county variation in predicted CPC trajectories that no linear function of the same covariates can replicate.

The panel structure provides additional identification leverage through dynamic over-identification (Bhargava, 1991; Liu, Mroz, & Van der Klaauw, 2010). With $K = 10$ time-varying covariates and $T = 30$ years, predetermination implies that every lag of each covariate is orthogonal to the current structural error, generating approximately $K \times (T - 1) \approx 290$ moment conditions simultaneously. In a static linear IV setting, exploiting these conditions would require constructing separate instruments from each lag of each covariate, creating a many-weak-instruments problem. The forward simulation aggregates all of this information into a single strong instrument through the structural hazard model, which uses the covariates' time-ordering and interactions to predict CPC trajectories (Angeles, Guilkey, & Mroz, 1998). This aggregation is the computational payoff of the simulation approach: instead of choosing among hundreds of individually weak instruments, the hazard model imposes an economically motivated functional form that combines them into one instrument with a first-stage $F$ of 170.

### 4.3 Identification

The instrument predicts CPC presence using predetermined county characteristics fed through the hazard model. 2SLS uses that prediction to estimate the causal effect. The identification argument rests on a single assumption about the information structure of the data, from which the exclusion restriction follows mechanically.

The core identifying assumption is:

[12]The linear second-stage controls $X_{c,t} \subset W_{c,t-1}$, including county and year fixed effects, entered linearly achieve $R^2 = 0.80$ (Table A9, Column 3), only 0.02 below the flexible specification (Column 4, $R^2 = 0.82$). First-stage power therefore comes from a quantitatively modest but statistically potent nonlinear component of the instrument (first-stage $F = 170$). Table A10 shows that the IV estimate is stable when different categories of time-varying controls are dropped from the second stage, providing an informal over-identification check.

**Predetermination Assumption.** The structural error $u_{c,t}$ has mean zero conditional on lagged observables, county and year fixed effects:

$$\mathbb{E}[u_{c,t} \mid W_{c,t-1}, \mu_c, \lambda_t] = 0. \tag{4}$$

This assumption has a transparent economic interpretation. County fixed effects absorb all time-invariant differences across counties, including geographic, cultural, and institutional factors that determine both baseline CPC presence and baseline fertility patterns. Year fixed effects absorb aggregate trends affecting all counties simultaneously, including national policy changes, macroeconomic conditions, and secular shifts in fertility norms. Lagged abortion rates and demographic covariates in $W_{c,t-1}$ absorb persistent county-specific demand dynamics that evolve predictably from one period to the next. What remains in $u_{c,t}$ are period-specific, county-specific fertility shocks that prior observables cannot forecast: an unexpected local economic downturn, an unanticipated change in clinic access, or an idiosyncratic shift in contraceptive behavior. The assumption states that these residual shocks are not forecastable from the same observable history that the hazard model uses to predict CPC openings.

Predetermination delivers the exclusion restriction without requiring any separate argument. Since the instrument converges to $\mathbb{E}[\text{CPC}_{c,t} \mid W_{c,t-1}]$ and is therefore measurable with respect to $W_{c,t-1}$, the law of iterated expectations implies:

$$\mathbb{E}[u_{c,t} \cdot \text{IV}_{c,t} \mid \mu_c, \lambda_t] = \mathbb{E}\Big[\text{IV}_{c,t} \cdot \underbrace{\mathbb{E}[u_{c,t} \mid W_{c,t-1}, \mu_c, \lambda_t]}_{=0}\Big] = 0. \tag{5}$$

This connects the identification strategy to the formula instruments framework of Borusyak et al. (2025), where instruments derived from predetermined covariates inherit their exogeneity from the information structure rather than from institutional arguments about channels. Because the instrument converges to the conditional expectation, the recentering adjustment proposed by Borusyak et al. (2025) is trivially satisfied; Appendix C verifies this.

The hazard model determines instrument strength and complier weighting, not the exclusion restriction. Misspecifying the hazard weakens the first stage but does not introduce bias as long as predetermination holds. The Monte Carlo validation (§4.6) confirms this: the hazard is deliberately

misspecified in all cells, yet the IV recovers the true parameter with bias below 0.07.

#### 4.3.1 Threats to Identification

Predetermination would fail if lagged county observables forecast current fertility shocks—for example, if CPC openings were anticipated and triggered behavioral responses before the CPC appeared, if CPC entry coincided with abortion clinic closures for reasons the instrument cannot capture, or if the instrument's functional form happened to correlate with nonlinear time trends. Four specific threats are addressed empirically, with full results in Appendix G.1.

*Anticipatory responses.* If organizations or individuals anticipate CPC openings and adjust behavior in advance, pre-treatment fertility trends would differ between treated and control counties. Event study estimates (Figure I) show flat pre-trends for abortion, birth, and pregnancy rates, with joint $F$-tests failing to reject the null ($F = 1.11$, $p = 0.359$ for abortion; $F = 0.88$, $p = 0.547$ for birth; $F = 1.00$, $p = 0.446$ for pregnancy). The flat pregnancy pre-trends are particularly relevant for the mechanism analysis: they confirm that treated and control counties were on parallel pregnancy trajectories before CPC entry, strengthening the interpretation of the post-treatment pregnancy null as a causal zero (§5.2).

*Correlated provider access changes.* If CPC entry coincides with abortion clinic closures, the instrument would capture supply-side effects rather than demand-side CPC persuasion. Restricting to counties always more than 30 miles from the nearest abortion provider yields a virtually identical estimate with a strong first stage ($F = 149$; Table **??**).

*Persistent unobservables.* If twice-lagged unobservables affect both current CPC presence (through the hazard model's use of lagged variables) and current fertility outcomes, predetermination would fail through a backdoor channel. Controlling for twice-lagged abortion rates attenuates the estimate but it remains significant, and the first stage remains strong ($F = 116$; Table **??**). The attenuation is consistent with some persistence in the outcome process, not with a violation of the exclusion restriction. This specification provides a lower bound on the causal effect under partial departures from predetermination; the range between the baseline and lagged-outcome-controlled estimates bounds the effect under varying degrees of outcome persistence (§5.4).

*Functional form dependence.* If the instrument's predictive power comes entirely from the polynomial time trends in the hazard model (which could mechanically correlate with outcome

trends), the identification would be fragile. Varying the polynomial order from 2 to 5 leaves the estimate virtually unchanged (Table **??**), confirming that the cross-sectional variation in county characteristics, not the time-series specification, drives the instrument.

### 4.4 Estimation

**First Stage:**

$$\text{CPC}_{c,t} = \pi_0 + \pi_1 \text{IV}_{c,t} + \pi_2' X_{c,t} + \mu_c + \lambda_t + v_{c,t} \tag{6}$$

**Second Stage:**

$$y_{c,t} = \alpha + \gamma \widehat{\text{CPC}}_{c,t} + \delta' X_{c,t} + \mu_c + \lambda_t + u_{c,t} \tag{7}$$

where $y_{c,t}$ denotes the log fertility outcome (abortion rate or birth rate).

Because the instrument is a function of predetermined variables, standard 2SLS standard errors remain valid regardless of estimation error in the hazard model (Amemiya, 1974).[13] Standard errors are clustered at the county level throughout; cluster-bootstrapped standard errors (resampling counties, re-estimating the hazard, and re-simulating within each draw) are similar across all specifications.

With a continuous instrument and continuous endogenous variable, 2SLS estimates a weighted average of marginal causal responses; the complier characteristics (§4.5) show that identification is driven by mid-sized counties with evangelical institutional presence.

The second-stage controls match the hazard model covariates (all lagged one year), with two differences: lagged abortion rates enter the hazard model but not the second stage (county fixed effects absorb the cross-sectional variation), and the hazard model uses polynomial time trends while the second stage uses year fixed effects (§4.2.1). Observations are weighted by the square root of female population for variance stabilization.

Under endogenous CPC placement, OLS is expected to understate the causal effect: CPCs locate where abortion demand is high for unobserved reasons, attenuating OLS toward zero. A

[13] See also Amemiya (1985, Chapter 8). Instruments have unrestricted degrees of freedom: any function of exogenous variables is a valid instrument, so replacing one such function with a nearby estimated function does not affect the asymptotic distribution.

Poisson control function endogeneity test confirms this pattern, and the 2SLS estimate exceeds OLS in absolute value across all specifications (§5.2).[14]

#### 4.4.1 Functional Form

The primary specifications use log-transformed outcomes in 2SLS. At the county-year and county-year-age level, fewer than 4 percent of abortion observations and fewer than 1 percent of birth observations are zero, so log is well-behaved. For demographic subgroup cells, zero-count shares rise substantially: 4–18 percent for race-age cells and 31–39 percent for marital-age cells. In these thin cells, log drops zeros, selecting out observations where treatment effects may be largest. I therefore use the Poisson control function estimator (Wooldridge, 2014) for all subgroup decompositions, with the subgroup-specific female population as the exposure variable.[15]

### 4.5 Complier Characteristics

With a continuous instrument and continuous endogenous variable, 2SLS estimates a weighted average of marginal causal responses. Counties where actual CPC presence deviates most from the simulated instrument contribute most to identification. Table A8 in Appendix E characterizes these counties. The top leverage quartile has somewhat lower baseline abortion rates (12.3 vs. 14.3 per 1,000), similar birth rates, and a higher birth-to-abortion ratio at ages 25–29 (7.1 vs. 5.7). These are predominantly mid-sized counties (10,000–30,000 women) in the 1990s and early 2000s with evangelical institutional presence; metro counties with 90,000+ women contribute little because per-capita CPC intensity is low.

### 4.6 Validation

Monte Carlo experiments validate the estimator under severe designed endogeneity. The design crosses three true treatment effects ($\beta \in \{-0.10, -0.20, -0.30\}$) with three data generation

[14] For the subgroup decompositions in §5, where thin cells make log 2SLS unreliable and the Poisson is the natural estimator, the absolute magnitudes are less informative than the cross-age and cross-group patterns, which are consistent across all functional forms.

[15] The Poisson CF follows Wooldridge (2014): a linear first stage recovers residuals, which enter as a control function in the Poisson second stage with population exposure. For the race decomposition, exposure is the race-age-specific female population. For the marital decomposition, exposure is the marital-status-specific female population derived from Census/ACS benchmarks, so that coefficients are semi-elasticities of the married or unmarried rate.

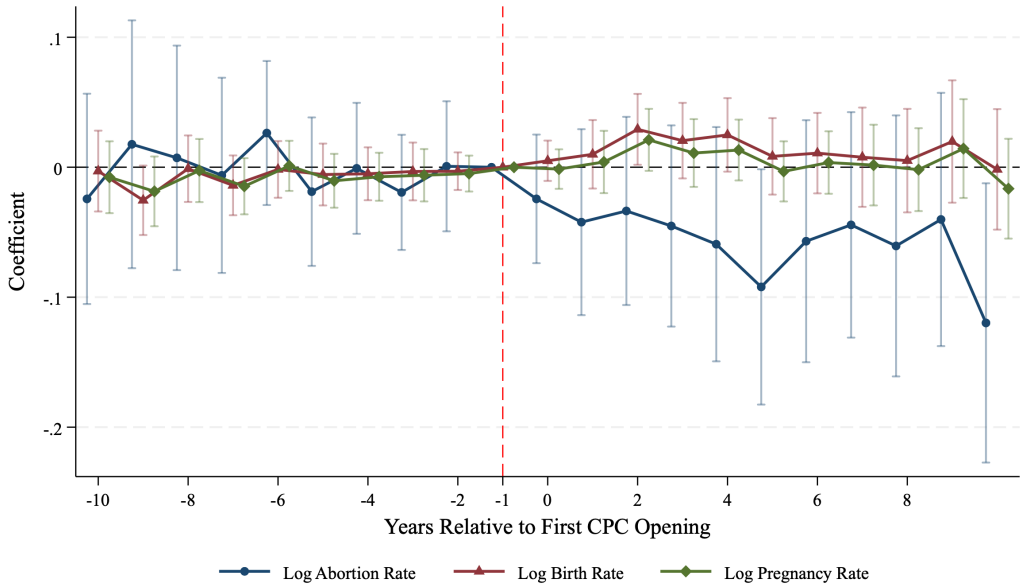
processes (DGP)—correct specification with an additive confounder, nonlinear confounding where the confounder enters through an interaction with treatment, and heterogeneous effects by county age composition—for a total of nine cells, each with 100 replications. The hazard model is deliberately misspecified in all cells: the true DGP includes confounders and functional forms not available to the hazard estimator, so the instrument is constructed from a model the researcher knows is wrong. OLS is severely biased in most cells, with mean estimates near +1.5 when the true effect is negative—a sign reversal that illustrates how badly endogenous CPC placement confounds the naive estimate. The IV recovers the true $\beta$ with bias below 0.07 in all nine cells (Table A6, Figure A4), despite the deliberate misspecification of the hazard model. This confirms that instrument validity depends on predetermination, not on the hazard model being correctly specified. Coverage of 95% confidence intervals is near nominal across the design. A permutation test shuffling the instrument across counties yields a first-stage $F$ of 0.69 (median) versus 170 in the actual data ($p = 0.000$; Appendix Table A7), confirming that instrument strength is not an artifact of overfitting county characteristics. Full details appear in Appendix D.

# 5 Results

## 5.1 Descriptive Evidence

Figure I presents event study estimates around first CPC opening. Abortion rate pre-trends show no systematic departure from zero, supporting the identifying assumption. Post-treatment abortion effects show reductions consistent with a CPC impact. Birth rate pre-trends are also flat. Pregnancy rate pre-trends show no differential trend before CPC entry, supporting the interpretation of the post-treatment pregnancy null as a causal zero rather than a continuation of pre-existing convergence (§5.2). IV estimates roughly double OLS across all specifications, confirming positive selection into CPC placement: CPCs locate where unobserved abortion demand is high, biasing OLS toward zero. The geographic expansion of CPCs from 1990 to 2019 is documented in Appendix Figures A2–A3; the staggered adoption pattern appears in Figure A1.

Figure I: Event Study: CPC Effects on Abortion, Birth, and Pregnancy Rates




*Notes*: Event study coefficients for log abortion rate, log birth rate, and log pregnancy rate relative to first CPC opening (event time = −1 is the omitted period). All specifications include county and year fixed effects, time-varying controls, and population weights. 95% confidence intervals based on standard errors clustered at the county level.

## 5.2 Main Results

Table II presents the main fertility estimates. The OLS estimate (Column 1) indicates that one additional CPC per 10,000 women reduces the abortion rate by 7.2%. The 2SLS estimate (Column 2) is roughly double at −14.4% (first-stage $F = 170$), consistent with positive selection bias: CPCs locate where abortion demand is high for unobserved reasons, attenuating OLS toward zero. The Poisson control function endogeneity test confirms this pattern; the CF residual is 0.111 ($p < 0.05$), rejecting the null of exogenous CPC placement. Across all specifications, the IV estimate exceeds OLS in absolute value.

The aggregate birth coefficient is positive but imprecisely estimated (0.019, s.e. 0.012). The aggregate pregnancy coefficient is also near zero (0.015, s.e. 0.012; Panel C), indicating that CPC presence does not reduce the rate at which women become pregnant. This null is informative about the mechanism. Under deterrence, where CPCs prevent pregnancies through upstream behavioral change, the pregnancy coefficient should be approximately −0.028 (applying the abortion reduction

to the pregnancy base rate).[16] The actual point estimate of +0.015 is roughly 3.6 standard errors from the deterrence prediction. The data are instead consistent with substitution: abortions fall, births rise by a similar amount (the +0.019 aggregate birth coefficient is of the right sign and order), and pregnancies are unaffected. The birth coefficient is imprecise because abortions are roughly 20 percent of pregnancies, so even full one-for-one substitution produces only a ∼3 percent increase in the birth rate, a small signal relative to the birth rate's cross-county variance.[17]

The pregnancy null disciplines the substitution accounting. The abortion rate is 11.1 per 1,000 women and the birth rate is 45.9 (Table I), so a 14.4% abortion reduction ($\approx$ 1.6 per 1,000) maps onto a $1.6/45.9 \approx 3.5\%$ birth increase under full one-for-one substitution. The actual birth coefficient of +1.9% is below full substitution but within the confidence interval, and the pregnancy null confirms that the discrepancy reflects imprecision rather than deterrence. Age-specific decompositions (§5.3) confirm that the pregnancy null holds across the age distribution.

[16] The mean abortion rate is 11.1 per 1,000 women; the mean pregnancy rate is 57.0. A 14.4% reduction in the abortion rate reduces pregnancies by $0.144 \times 11.1/57.0 \approx 0.028$ in log points, if no offsetting birth increase occurs.

[17] The abortion panel shows $N = 4{,}376$ while the birth and pregnancy panels show $N = 4{,}380$. The difference reflects four county-year observations with zero abortion counts (Clay, Graham, Perquimans, and Yadkin counties, each in a single year) that are dropped by the log transformation. All 146 counties remain in the panel.

Table II: Effect of Crisis Pregnancy Centers on Abortion, Birth, and Pregnancy Rates

| | (1) OLS | (2) 2SLS | (3) Poisson-CF |
|---|---|---|---|
| *First Stage: CPCs per 10,000 on Simulated Instrument* | | | |
| Simulated CPCs per 10,000 | | 0.838*** | |
| | | ( 0.064) | |
| *Panel A: Log Abortion Rate* | | | |
| CPCs per 10,000 | -0.072** | -0.144*** | -0.075* |
| | ( 0.029) | ( 0.040) | ( 0.041) |
| CF residual (endog. test) | | | 0.111** |
| | | | ( 0.048) |
| N | 4,376 | 4,376 | 4,380 |
| First-stage F | | 170.2 | |
| *Panel B: Log Birth Rate* | | | |
| CPCs per 10,000 | 0.010 | 0.019 | 0.020 |
| | ( 0.008) | ( 0.012) | ( 0.017) |
| CF residual (endog. test) | | | -0.017 |
| | | | ( 0.019) |
| N | 4,380 | 4,380 | 4,380 |
| First-stage F | | 170.3 | |
| *Panel C: Log Pregnancy Rate* | | | |
| CPCs per 10,000 | 0.010 | 0.015 | 0.025 |
| | ( 0.008) | ( 0.012) | ( 0.018) |
| CF residual (endog. test) | | | -0.016 |
| | | | ( 0.020) |
| N | 4,380 | 4,380 | 4,380 |
| First-stage F | | 170.3 | |
| County FE | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes |
| $\sqrt{\text{pop}}$ weights | Yes | Yes | — |

*Notes:* Standard errors clustered at the county level in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Column (1): OLS with county and year fixed effects. Column (2): 2SLS using simulated CPC instrument. Column (3): Poisson regression with control function (Wooldridge, 2014); exposure = county female population; analytic SEs. A specification jointly instrumenting CPC count and distance to nearest CPC (both endogenous regressors instrumented) yields a larger abortion coefficient ($-0.265$) but with a first-stage $F$ of 2.4 (available on request). Pregnancy rate equals abortion rate plus birth rate plus fetal deaths per 1,000 women. All specifications include county demographic controls. First-stage $F$ differs slightly across panels because the first stage is re-estimated on each panel's sample ($N$ = 4,376 for log abortion, 4,380 for log birth and pregnancy).

### 5.3 Age-Specific Results

Table III decomposes CPC effects by age group using separate county-year 2SLS regressions (first-stage $F > 127$).[18] The abortion reduction is pervasive: coefficients are negative and significant for all age groups, ranging from $-0.136$ (ages 35–44) to $-0.212$ (ages 25–29). Birth coefficients are positive and significant at ages 25 and older, with small magnitudes consistent with the arithmetic of substitution (the birth base rate is roughly five times the abortion rate). All five pregnancy coefficients are statistically indistinguishable from zero, confirming that CPC-induced abortion reductions reflect substitution rather than deterrence at every age.

An exploratory marital decomposition (Appendix F.4) asks where the substituting births land across demographic subgroups. Age-specific substitution ratios (Appendix Table A18) are consistent with one-for-one substitution at ages 25–29 (ratio = 0.93, 95% CI includes unity) and above unity at older ages, though imprecisely estimated.

Figure II summarizes the age-specific coefficients for abortion, birth, and pregnancy effects.

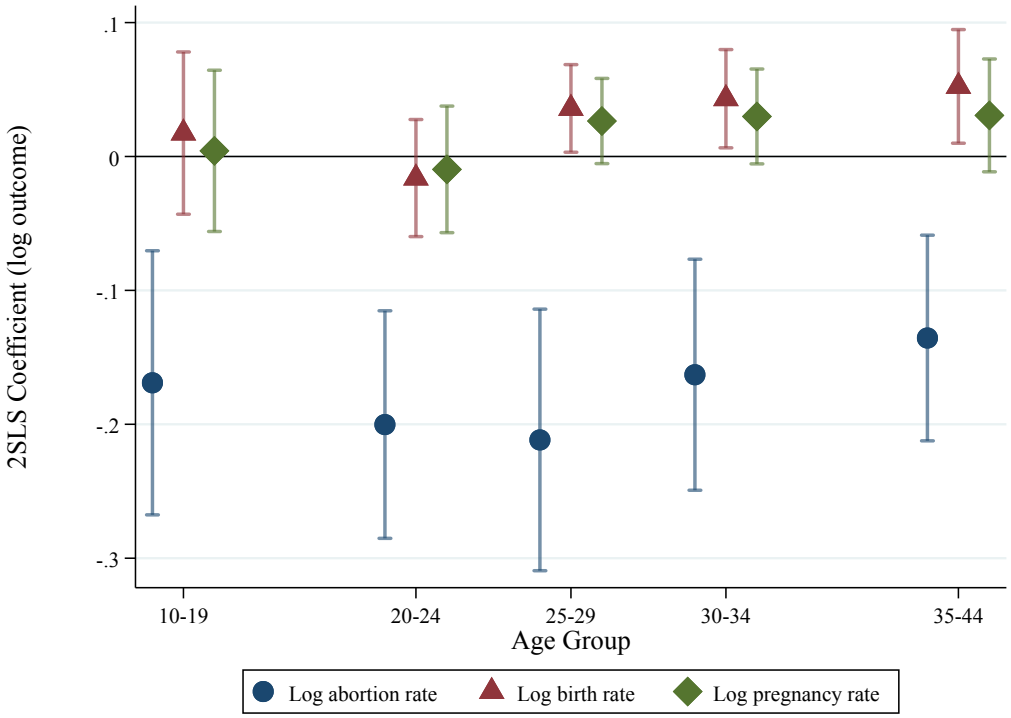
[18] Treatment is measured as lagged CPCs per 10,000 women using the county-level total female population as the denominator across all age groups, ensuring a common treatment measure since CPC services are not age-restricted. A pooled specification with age × year fixed effects (AYFE) produces similar abortion coefficients (Appendix Table A11), confirming that the age gradient is not driven by confounding from age-specific national trends.

Table III: Age Heterogeneity in CPC Effects (2SLS)

| | (1) 10–19 | (2) 20–24 | (3) 25–29 | (4) 30–34 | (5) 35–44 |
|---|---|---|---|---|---|
| *Panel A: Log Abortion Rate* | | | | | |
| CPCs per capita | -0.169*** | -0.200*** | -0.212*** | -0.163*** | -0.136*** |
| | ( 0.050) | ( 0.043) | ( 0.050) | ( 0.044) | ( 0.039) |
| N | 4,380 | 4,380 | 4,380 | 4,380 | 4,380 |
| First-stage F | 127.5 | 135.4 | 132.0 | 132.3 | 137.3 |
| *Panel B: Log Birth Rate* | | | | | |
| CPCs per capita | 0.017 | -0.016 | 0.036** | 0.043** | 0.052** |
| | ( 0.031) | ( 0.022) | ( 0.017) | ( 0.019) | ( 0.022) |
| N | 4,380 | 4,380 | 4,380 | 4,380 | 4,380 |
| First-stage F | 127.5 | 135.4 | 132.0 | 132.3 | 137.3 |
| *Panel C: Log Pregnancy Rate* | | | | | |
| CPCs per capita | 0.004 | -0.010 | 0.026 | 0.030* | 0.031 |
| | ( 0.031) | ( 0.024) | ( 0.016) | ( 0.018) | ( 0.021) |
| N | 4,380 | 4,380 | 4,380 | 4,380 | 4,380 |
| First-stage F | 127.5 | 135.4 | 132.0 | 132.3 | 137.3 |
| County FE | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes | Yes | Yes |

*Notes:* Each column estimates the 2SLS specification from Table II Column (2) separately for each age group. Standard errors clustered at the county level. Data at the county-year-age level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure II: CPC Effects by Age Group: Abortion, Birth, and Pregnancy Rates (2SLS)




*Notes*: Point estimates and 95% confidence intervals from the separate age-group 2SLS specifications in Table III. Abortion coefficients (circles) are negative and significant across all age groups. Birth coefficients (triangles) are positive and significant at ages 25 and older. Pregnancy coefficients (diamonds) cluster at zero across all age groups, consistent with substitution rather than deterrence (§5.3).

### 5.4 Robustness

The abortion result is stable across identification tests, alternative estimators, and potential confounders. Table IV summarizes 13 specifications; full results appear in the referenced appendix tables. Three features merit emphasis.

Table IV: Robustness Summary: Log Abortion Rate

| Specification | $\hat{\beta}$ | SE | First-stage F | N |
|---|---|---|---|---|
| Baseline 2SLS | -0.144*** | ( 0.040) | 170.2 | 4,376 |
| OLS | -0.072** | ( 0.029) | — | 4,376 |
| TRAP + PI law controls | -0.137*** | ( 0.038) | 168.9 | 4,376 |
| Always far from provider (>30mi) | -0.152*** | ( 0.044) | 149.0 | 3,027 |
| Lagged outcome control | -0.098*** | ( 0.034) | 115.9 | 4,080 |
| $X_c\times$ year FE | -0.130*** | ( 0.036) | 163.6 | 4,376 |
| + Prot×Rep×year interaction | -0.133*** | ( 0.036) | 161.6 | 4,376 |
| Pre-2009 subsample | -0.172*** | ( 0.061) | 45.4 | 2,773 |
| Drop border counties | -0.178*** | ( 0.056) | 90.7 | 3,387 |
| NC only | -0.167*** | ( 0.041) | 96.6 | 2,996 |
| SC only | -0.077 | ( 0.072) | 74.9 | 1,380 |
| Poisson-CF | -0.075* | ( 0.041) | — | 4,380 |
| Unweighted 2SLS | -0.137*** | ( 0.039) | 163.2 | 4,376 |

*Notes:* Each row re-estimates the main 2SLS specification with one modification. The dependent variable is log county abortion rate throughout. All specifications include county and year fixed effects and the full control set unless noted otherwise. Population-weighted unless noted. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The credible range for the abortion coefficient is bounded by the trending-confounder specification ($-0.130$, adding $X_c\times$ year FE interactions; Table A19) and the baseline ($-0.144$). Controlling for twice-lagged abortion rates provides a lower bound of $-0.098$ under partial departures from predetermination (Table **??**). All three estimates are significant at the 1% level with first-stage $F > 100$.

Joint $F$-tests of pre-treatment event study coefficients fail to reject zero pre-trends for abortion ($p = 0.359$), birth ($p = 0.547$), and pregnancy ($p = 0.446$). The flat pregnancy pre-trends rule out the possibility that the post-treatment pregnancy null reflects pre-existing convergence. A leave-two-county-out jackknife shows no pair shifts the estimate by more than 20% (Figure A11), and a permutation test confirms instrument strength is not an artifact of overfitting ($p = 0.000$;

Table A7). Oster (2019) reduced-form $\delta^* = 10.7$ for abortion (Appendix G.10), and Conley, Hansen, and Rossi (2012) bounds exclude zero for direct instrument effects up to $|\delta| = 0.05$ (Table A27).

The Poisson CF, which does not impose log linearity and retains zero-count observations, yields $-0.075$ (Table II). The gap between the 2SLS and Poisson CF estimates reflects differences in functional form assumptions and weighting rather than a failure of identification: 2SLS is the more familiar estimator, directly comparable to OLS, and does not require the conditional mean assumptions of the Poisson; augmented synthetic control estimators produce consistent negative ATTs (Appendix H), and the IV-implied binary ATT closely matches the SCPI estimate (Appendix Table A30). The range $-0.075$ to $-0.144$ brackets the effect across estimators that make different assumptions about the outcome distribution.

# 6 Discussion

## 6.1 Aggregate Magnitudes

CPC intensity increased from a mean of 0.21 per 10,000 women in 1990 to 0.40 in 2019. Using the 2SLS estimate, CPC expansion reduced county abortion rates by roughly 3 percent over this period. County-level abortion rates fell 49% between 1990 and 2019; CPC expansion accounts for approximately 6% of this aggregate decline using the 2SLS LATE (which reflects complier counties and likely overstates the population-average effect), or roughly 3% using the Poisson CF estimate.[19] Counties with above-median CPC funding show larger abortion reductions than those with below-median funding ($-0.50$ vs. $-0.20$; Appendix Table A12), suggestive of the effect scaling with organizational resources. The first-stage $F$ in the high-funding subsample is 8.0, however, so the magnitudes should be treated as descriptive rather than well-identified IV estimates.

## 6.2 Welfare Implications

The welfare implications depend on whether the marginal births produced by CPC-induced substitution are wanted or unwanted, a question the data cannot answer. Because CPCs do not

[19] For comparison, Lindo et al. (2020) find that clinic closures reduced abortion rates by approximately 20% in affected areas. The baseline CPC estimate ($-0.144$) is of similar order to the supply-side estimates; the conservative $X_c\times$ year FE specification yields $-0.130$ (Table IV). Reconciliation of the IV and synthetic control estimates appears in Appendix Table A30.

prevent pregnancies, the relevant comparison is between the welfare cost of an abortion and the welfare cost of carrying to term for the marginal woman who is redirected. The exploratory marital decomposition (Appendix F.4) provides partial guidance: the births that appear in the data concentrate among married women aged 25–34, a group for whom carrying to term is plausibly close to desired fertility timing. Unmarried teenagers show the other clean substitution cell, and for that group the welfare calculus is less favorable. Linking CPC exposure to long-run maternal and child outcomes would clarify these calculations but requires individual-level data that are currently unavailable.

### 6.3 Limitations

The pregnancy null establishes substitution as the aggregate mechanism. The marital decomposition (Appendix F.4) characterizes where births land but has important limitations, including endogenous marital status and interpolated denominators. For the age groups where the married birth increases are largest (25–29 and 30–34), a marriage balance test finds no evidence of CPC-induced reclassification across marital status (Appendix Table A16). Power calculations (Appendix Table A17) confirm that the "missing" unmarried births at ages 20–29 are genuinely absent rather than hidden by imprecision. The county-level data cannot distinguish within-cell substitution from cross-demographic spillovers, and a structural model pinning down mechanisms would require individual-level data on women's pathways through CPC contact, pregnancy, and fertility decisions.

### 6.4 External Validity and the Post-*Dobbs* Environment

These estimates are identified from two southeastern states with specific evangelical Protestant demographics, a particular regulatory environment, and a CPC network that expanded during a specific historical window. The IV is most informative for mid-sized counties (10,000–30,000 women) that experienced meaningful CPC expansion between 1990 and the 2010s. Generalizability to other regions or time periods is unknown.

The post-*Dobbs* environment presents a qualitatively different setting. During the study period, abortion was constitutionally protected and relatively accessible in both NC and SC; CPCs operated as demand-side influences competing with legal abortion access. The estimates capture the

effect of CPCs as persuasive institutions in a setting where women had a meaningful outside option. In states with near-total abortion bans, CPCs are no longer competing with accessible abortion services. Their role may have shifted from persuading women to forgo abortions to supporting those who face legal or financial barriers to obtaining one. Whether the demand-side effects identified here persist under these new constraints is an open question with direct policy relevance, given that many restrictive states have substantially increased public CPC funding since *Dobbs*.

# 7 Conclusion

CPCs reduce abortion rates by an estimated 13–14 percent per unit of CPC intensity. The abortion result is robust across functional forms, weighting schemes, and alternative estimators, with a first-stage $F$ above 100 in every county-level specification and above 20 in the pooled age-group specifications. This is the paper's primary contribution: the first causal estimate of CPC effects on fertility.

The pregnancy null rules out deterrence as the primary channel: pregnancy rates do not respond to CPC presence at any age group or specification, so the abortion reductions reflect changes in how pregnancies are resolved, not whether they occur. An exploratory marital decomposition (Appendix F.4) reveals that abortion reductions are pervasive among unmarried women across all adult ages, while the corresponding birth increases concentrate among married women at ages 25–34. The demographic heterogeneity is suggestive, though the county-level data cannot determine whether the same women appear in both cells or whether cross-demographic spillovers are at work.

These findings have direct implications for the post-*Dobbs* environment, where many states have substantially increased CPC funding. If CPCs work by changing how pregnancies are resolved rather than whether they occur, their effectiveness depends on the availability of abortion as an outside option. In states with near-total abortion bans, the demand-side persuasion mechanism identified here may have less scope to operate. Future work could link CPC exposure to long-run maternal and child outcomes, examine interactions with public programs such as Medicaid and paid family leave, and exploit individual-level CPC utilization data to distinguish pathways.

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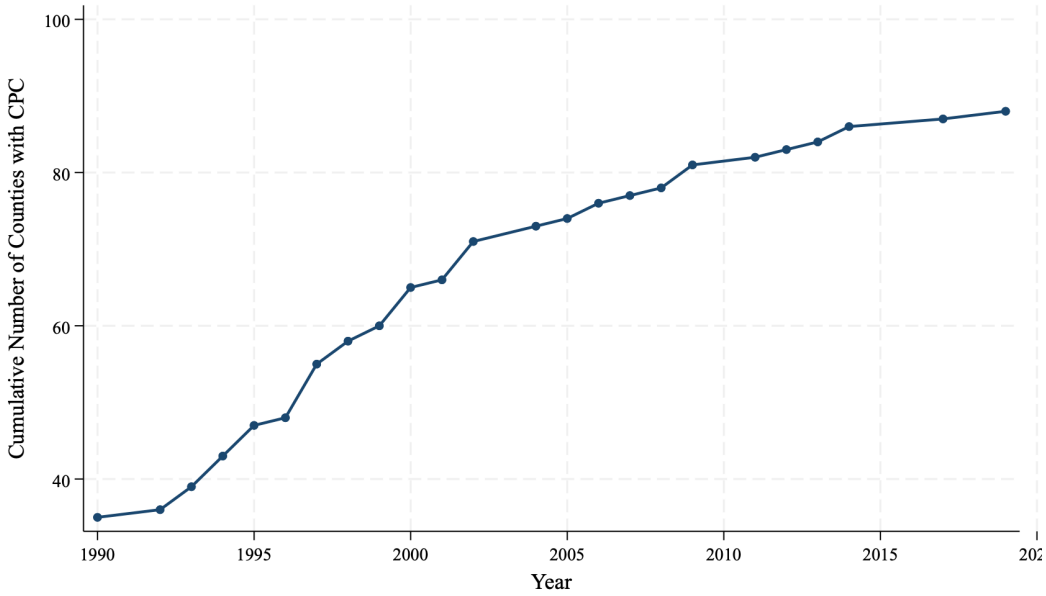
# A Institutional Background

Figure A1: CPC Expansion: Staggered Adoption Across Counties



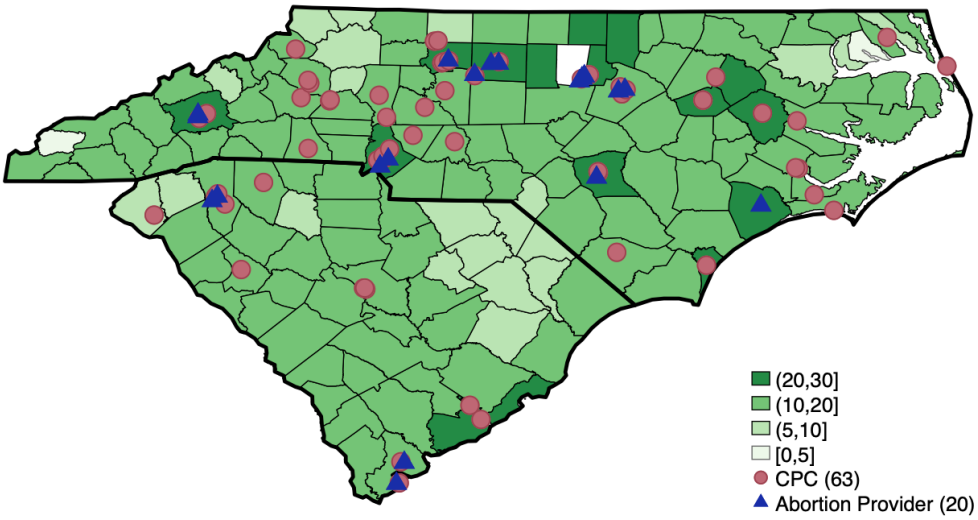

*Notes*: Distribution of first CPC opening year across NC and SC counties.

Table A1: CPC Public Funding by State

| State | Annual Public Funding (approx.) |
|---|---:|
| Texas | $100,000,000 |
| Pennsylvania | $8,200,000 |
| Florida | $4,500,000 |
| Ohio | $7,500,000 |
| Oklahoma | $5,000,000 |
| North Dakota | $3,500,000 |
| Michigan | $3,300,000 |

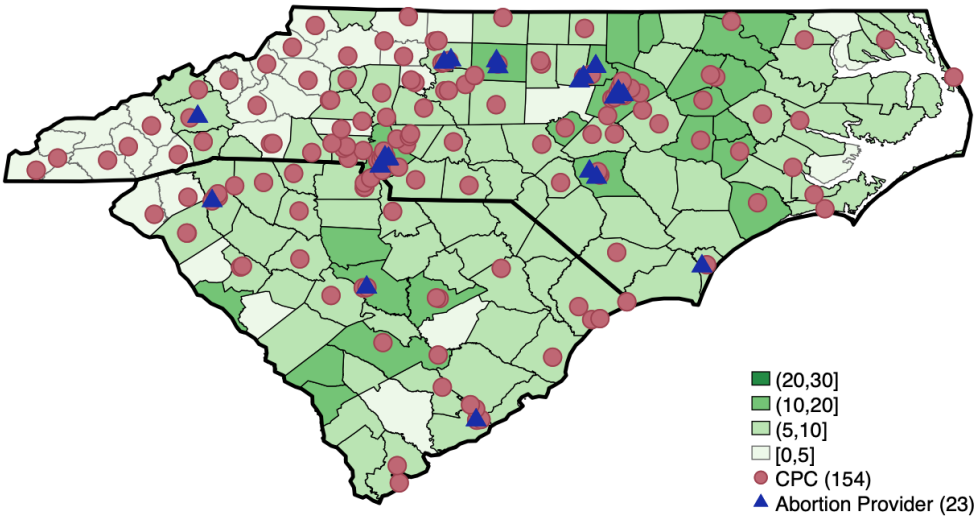
Source: State budgets and health departments via Associated Press report, "Millions in tax dollars flow to anti-abortion centers in US." Kimberlee Kruesi. 02/05/2022.

Figure A2: CPCs, Abortion Providers and Abortion Rate (NC & SC): 1990




*Notes*: Each dot represents a CPC (triangle) or abortion provider (circle). County shading indicates the abortion rate per 1,000 women aged 10–44.

Figure A3: CPCs, Abortion Providers and Abortion Rate (NC & SC): 2019




*Notes*: Each dot represents a CPC (triangle) or abortion provider (circle). County shading indicates the abortion rate per 1,000 women aged 10–44.

distribution as a sequence of independent binary events, each estimable by logit regardless of the true count-generating process. State dependence enters through $\text{CPC}_{c,t-1}$, so the hazard at each step conditions on accumulated stock. The outcome $\text{NewCPC}_{c,t}$ resets to zero each year and can equal one multiple times for the same county across $t$, with the likelihood taking the standard pooled logit form:

$$\mathcal{L} = \prod_c \prod_t p_{c,t}^{\text{NewCPC}_{c,t}} (1 - p_{c,t})^{1-\text{NewCPC}_{c,t}} \tag{8}$$

This structure allows the hazard model to capture both first entries and subsequent CPC additions, with the existing stock entering as a time-varying covariate that governs the rate of accumulation.

### B.3 Forward Simulation Algorithm

For each county $c$ and target year $t$, I generate $S = 1{,}000$ simulated CPC expansion paths:

1. **Initialize:** Set $\widehat{\text{CPC}}_{c,1990}^{(s)} = \text{CPC}_{c,1990}$ (observed initial state) for all $s$.
2. **Forward simulate:** For each draw $s = 1, \ldots, S$ and year $\tau = 1991, \ldots, t$:
   (a) Compute predicted opening probability $\widehat{p}_{c\tau}^{(s)}$ using simulated lagged CPC counts, distances, and time trends, with all other covariates held at their observed values;
   (b) Draw $U_{c\tau}^{(s)} \sim \text{Uniform}(0, 1)$;
   (c) Open a CPC if $U_{c\tau}^{(s)} \leq \widehat{p}_{c\tau}^{(s)}$;
   (d) Update cumulative count, distance to nearest CPC, and polynomial time trend.

### B.4 Alternative Hazard Specifications

Table A4 re-estimates the IV using an alternative hazard specification that replaces the polynomial time trend with year fixed effects and drops lagged abortion rates from the predictor set. The abortion coefficient is virtually unchanged ($-0.135$ to $-0.145$ vs. baseline $-0.144$; Table A4).

Table A3: Variable Treatment in the Forward Simulation

| Category | Variables | Rationale |
|---|---|---|
| *Updated within simulation* | CPC count, distance to nearest CPC, polynomial time trend | Endogenous to simulated expansion process |
| *Held at observed values* | Demographics, religious composition, political ideology, unemployment, distance to nearest abortion provider | Evolve in data but treated as exogenous to CPC openings within each draw |
| *Frozen at 1990* | Lagged abortion rates | Avoids embedding assumed treatment effects |

*Notes:* "Updated" variables change recursively within each of the $S = 1{,}000$ simulation draws as new CPCs open. "Observed" variables enter at their actual panel values in each county-year. "Frozen" variables are fixed at 1990 cross-sectional values throughout the simulation.

Table A4: Sensitivity to Alternative Hazard Model Specifications

| | County Aggregate | | | | Age 25-29 Separate | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) Baseline | (2) No abort | (3) No religion | (4) Cubic | (5) Baseline | (6) No abort | (7) No religion | (8) Cubic |
| CPCs per 10,000 | -0.144*** | -0.135*** | -0.145*** | -0.142*** | -0.238*** | -0.215** | -0.241*** | -0.232*** |
| | (0.040) | (0.039) | (0.041) | (0.039) | (0.088) | (0.088) | (0.089) | (0.085) |
| N | 4,376 | 4,376 | 4,376 | 4,376 | 4,302 | 4,302 | 4,302 | 4,302 |
| First-stage F | 170.2 | 155.4 | 174.1 | 189.1 | 134.2 | 130.5 | 140.1 | 142.1 |

*Notes:* Each column re-estimates the full pipeline (hazard model, forward simulation, 2SLS) under an alternative hazard specification. County and year FE, controls, $\sqrt{\text{pop}}$ weights. Standard errors clustered at county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Sensitivity to Abortion Rate Updating in Simulation

| | Log Abortion Rate | | Log Birth Rate | |
|---|---|---|---|---|
| CPCs per 10,000 | -0.152*** | -0.138*** | 0.020 | 0.021 |
| | (0.039) | (0.038) | (0.013) | (0.013) |
| N | 4,376 | 4,376 | 4,380 | 4,380 |
| First-stage F | 129.3 | 108.5 | 129.4 | 108.5 |

*Notes:* Odd columns use the baseline instrument with lagged abortion rates evolving via an auxiliary model during the forward simulation. Even columns use an alternative instrument with lagged abortion rates frozen at 1990 observed values. County and year FE, $\sqrt{\text{pop}}$ weights, standard controls. Standard errors clustered at county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# C Formal Identification

## C.1 Two-Assumption Decomposition

The main text's predetermination assumption can be decomposed into two conditions.

**Assumption 1 (Conditional Orthogonality):** Actual organizational timing shocks are mean-independent of unobserved fertility shocks, conditional on observable history and fixed effects:

$$\varepsilon_{c,t}^{\mathrm{DGP}} \perp u_{c,t} \mid \mathcal{F}_{c,t-1}, \mu_c, \lambda_t \tag{9}$$

where $\mathcal{F}_{c,t-1} = \{W_{c,\tau}\}_{\tau=1}^{t-1}$ denotes the historical information set.

**Assumption 2 (Predetermination):** Observable county characteristics are predetermined with respect to current-period fertility shocks:

$$\mathbb{E}[u_{c,t} \mid W_{c,t-1}, \mu_c, \lambda_t] = 0. \tag{10}$$

Since the instrument converges to $\mathbb{E}[\mathrm{CPC}_{c,t} \mid W_{c,t-1}]$ and is measurable with respect to $W_{c,t-1}$, predetermination directly establishes the exclusion restriction via the law of iterated expectations.

# D Monte Carlo Validation

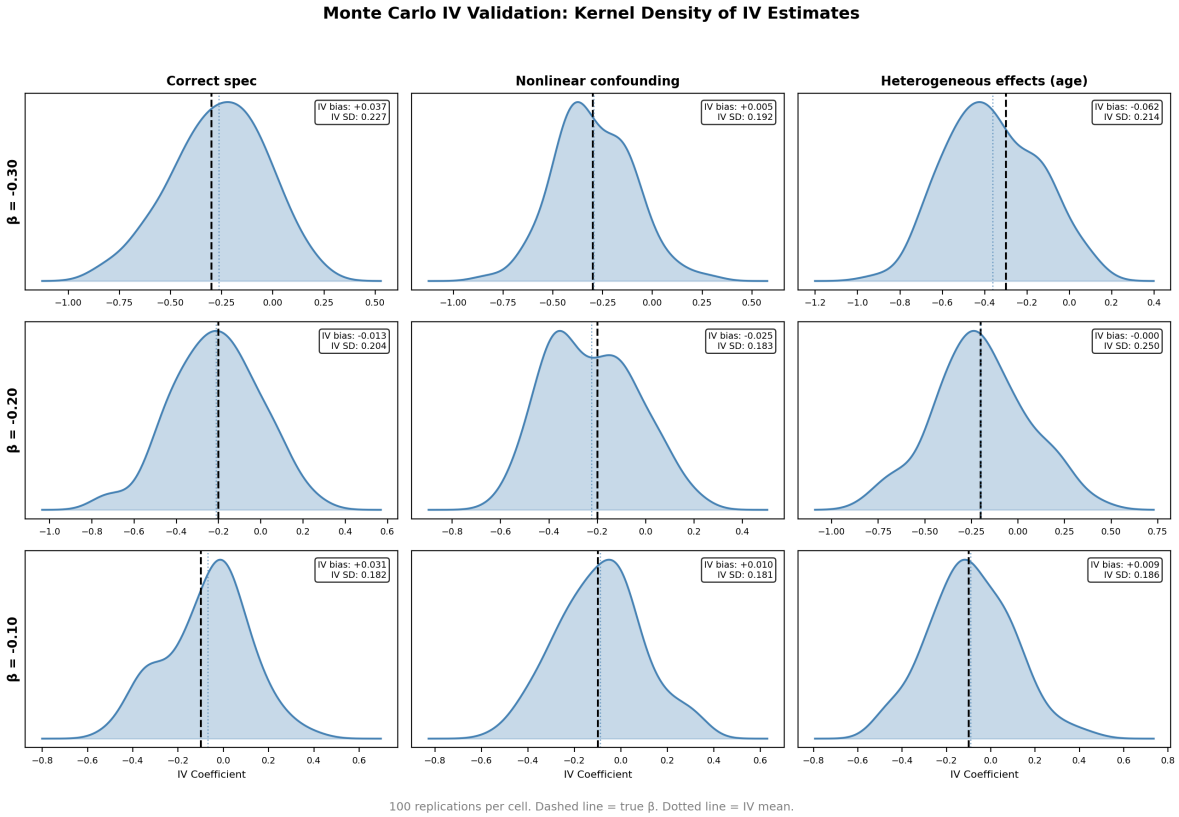
The $3 \times 3$ factorial design crosses three true treatment effects ($\beta \in \{-0.10, -0.20, -0.30\}$) with three DGP specifications (correct specification with additive confounder; nonlinear confounding; heterogeneous effects by county age composition), each repeated for 100 replications. The hazard model is deliberately misspecified.

Table A6: Monte Carlo Validation: Summary Statistics Across DGP Specifications

| True $\beta$ | DGP | Mean OLS | Mean IV | Bias OLS | Bias IV | SD(IV) | Mean $F$ | Reps |
|---|---|---|---|---|---|---|---|---|
| $-0.30$ | Correct spec | +1.444 | -0.263 | +1.744 | +0.037 | 0.228 | 74 | 100 |
| | Nonlinear confounding | -0.231 | -0.295 | +0.069 | +0.005 | 0.193 | 75 | 100 |
| | Heterogeneous effects (age) | +1.434 | -0.362 | +1.734 | -0.062 | 0.215 | 74 | 100 |
| $-0.20$ | Correct spec | +1.518 | -0.213 | +1.718 | -0.013 | 0.205 | 74 | 100 |
| | Nonlinear confounding | -0.167 | -0.225 | +0.033 | -0.025 | 0.183 | 74 | 100 |
| | Heterogeneous effects (age) | +1.528 | -0.200 | +1.728 | -0.000 | 0.252 | 74 | 100 |
| $-0.10$ | Correct spec | +1.611 | -0.069 | +1.711 | +0.031 | 0.183 | 74 | 100 |
| | Nonlinear confounding | -0.088 | -0.090 | +0.012 | +0.010 | 0.182 | 74 | 100 |
| | Heterogeneous effects (age) | +1.621 | -0.091 | +1.721 | +0.009 | 0.187 | 74 | 100 |

*Notes:* Each cell reports summary statistics from Monte Carlo replications. True $\beta$ is the known treatment effect in the synthetic DGP. "Correct spec" uses linear, additive confounding. "Nonlinear confounding" replaces $Z$ with $Z^2$. "Heterogeneous effects (age)" introduces treatment effect heterogeneity scaled by county age composition (share aged 20–29). Mean OLS uses the confounded treatment variable; Mean IV uses the forward-simulation instrument. Bias = Mean estimate − True $\beta$. $F$ is the first-stage Kleibergen-Paap $F$-statistic.

Figure A4: Monte Carlo IV Validation: Kernel Density of IV Estimates

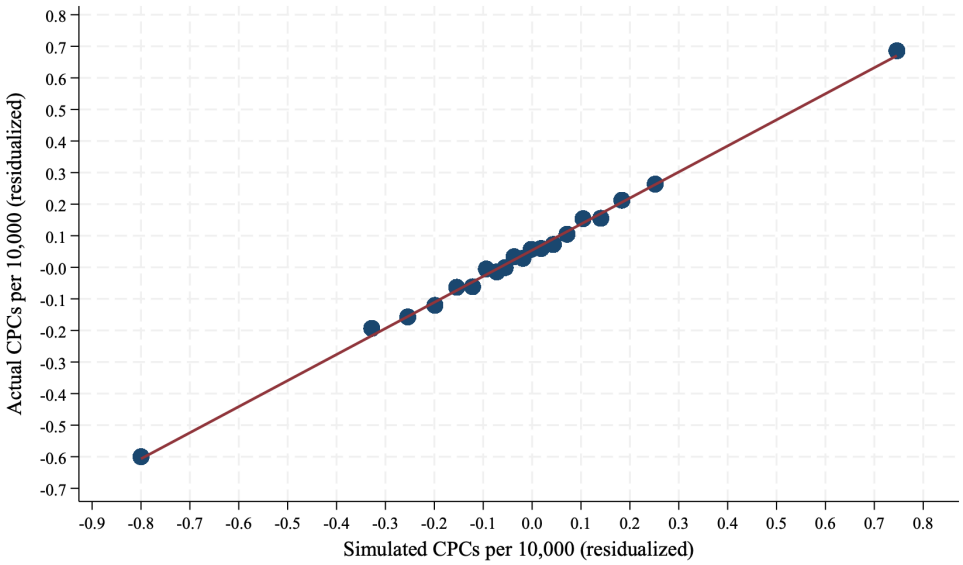



*Notes*: Kernel density of IV coefficient estimates across the $3 \times 3$ Monte Carlo design (100 replications per cell). Dashed black line marks the true $\beta$; dotted blue line marks the IV mean.

# E First Stage Diagnostics

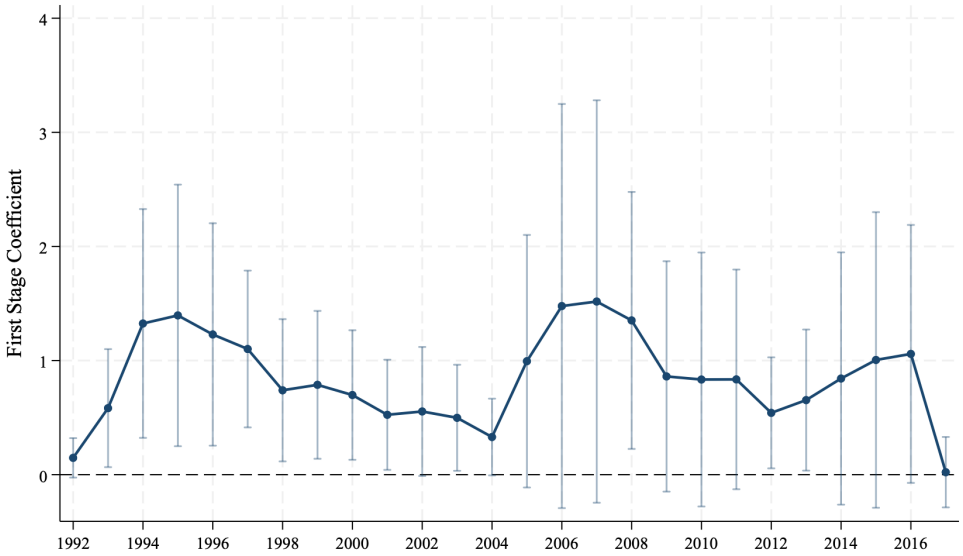
Figure A5 plots actual against simulated CPC intensity after residualizing on county and year fixed effects, confirming a strong positive relationship. Figure A6 reports rolling 5-year-window first-stage coefficients across the sample period. The instrument is strongest in the 1990s and early 2000s, when CPC expansion was most active; coefficients weaken and confidence intervals widen in post-2010 windows as CPC entry slows and the remaining variation becomes less informative. This temporal concentration is consistent with the larger reduced-form estimate in the pre-2009 subsample ($-0.172$, $F = 45.4$; Table IV) and supports interpreting the main results as primarily identified from the earlier expansion period.

Figure A5: First Stage: Actual vs. Simulated CPC Intensity




*Notes*: Binscatter of actual CPC intensity against simulated CPC intensity. Both variables residualized on county and year fixed effects and controls.

Figure A6: Rolling-Window First-Stage Coefficients




*Notes*: First-stage coefficients from rolling 5-year windows across the sample period, with 95% confidence intervals.

Table A7: Shuffled Instrument Permutation Test

| | First-stage $F$ | Reduced-form $\hat{\pi}$ |
|---|---|---|
| Actual instrument | 170.2 | -0.1150 |
| | | ( 0.0345) |
| Shuffled median | 0.686 | -0.0032 |
| Shuffled mean | 1.649 | -0.0027 |
| Shuffled SD | | 0.0272 |
| Shuffled 95th pctile | 6.953 | |
| Permutation $p$-value | 0.0000 | 0.0000 |

*Notes:* 500 permutations randomly reassign county labels to instrument paths, preserving the time-series structure. Permutation $p$-value is the fraction of shuffled statistics exceeding the actual value.

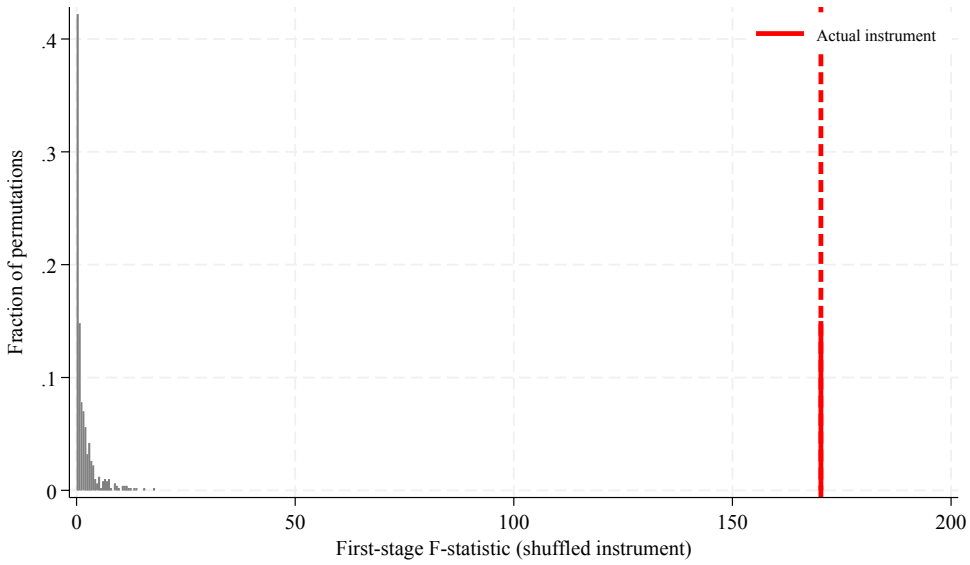



Figure A7: Permutation Distribution of First-Stage $F$-Statistic

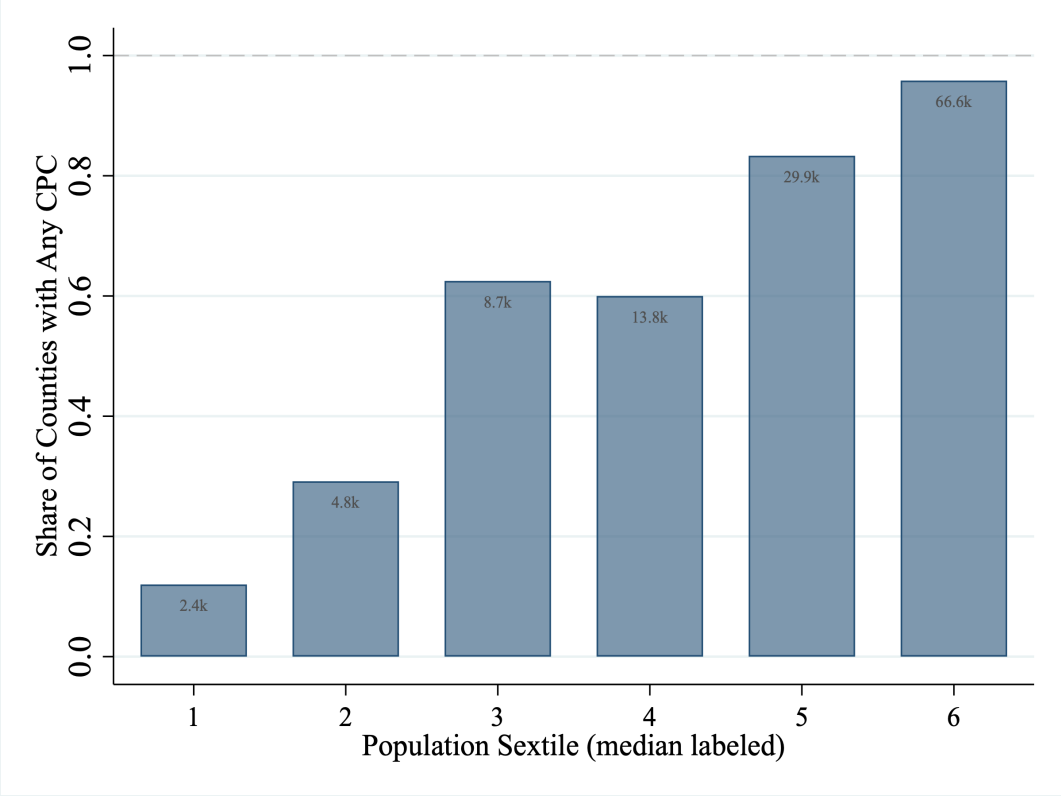
*Notes:* Histogram of first-stage $F$-statistics from 500 permutations that randomly reassign county labels to instrument paths. The dashed red line marks the actual first-stage $F = 170$.

Table A8: Complier County Baseline Characteristics (1990)

| | COMP | | | |
|---|---|---|---|---|
| | Full Sample | Top Leverage Q | Bottom 3 Q | p-value |
| Aggregate Abortion Rate | 13.81 | 12.25 | 14.32 | 0.005 |
| Aggregate Birth Rate | 50.46 | 49.52 | 50.77 | 0.553 |
| Aggregate Birth/Abortion | 4.10 | 4.25 | 4.04 | 0.493 |
| Age 25-29 Abortion Rate | 15.82 | 13.70 | 16.52 | 0.028 |
| Age 25-29 Birth Rate | 83.31 | 87.93 | 81.80 | 0.395 |
| Age 25-29 Birth/Abortion | 6.04 | 7.12 | 5.68 | 0.079 |
| Age 20-24 Abortion Rate | 29.19 | 26.22 | 30.17 | 0.108 |
| Age 20-24 Birth Rate | 112.74 | 120.41 | 110.23 | 0.177 |
| Age 20-24 Birth/Abortion | 4.94 | 5.42 | 4.79 | 0.414 |

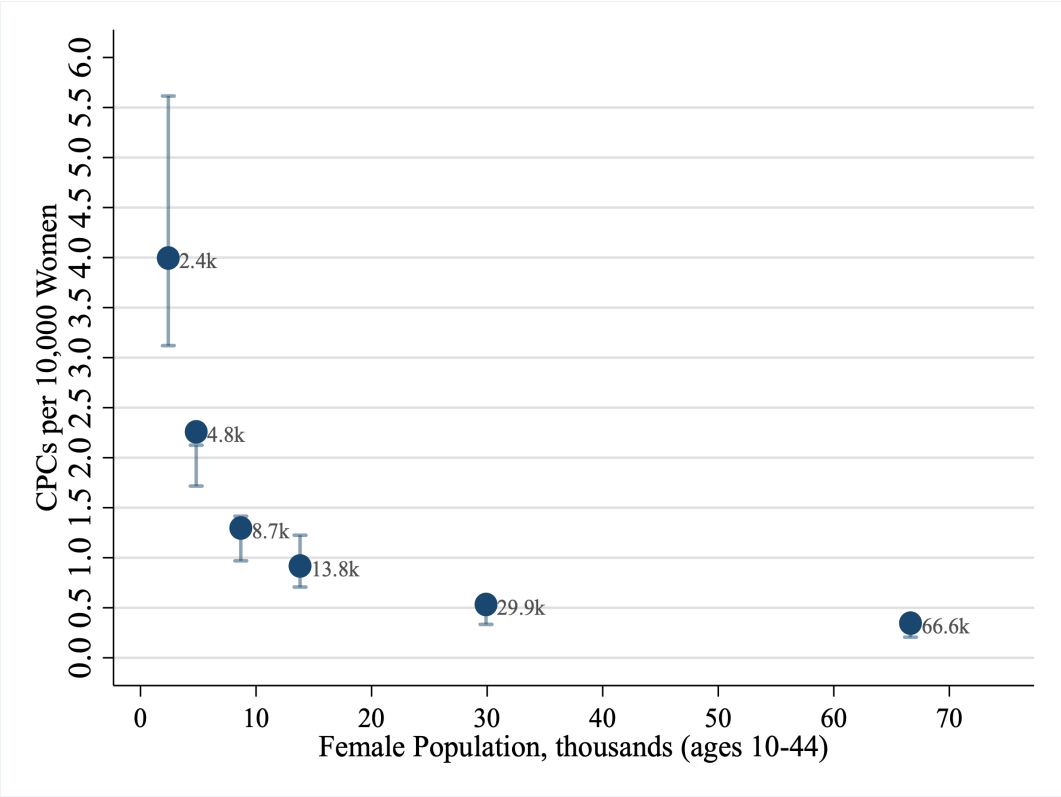
*Notes:* Leverage quartiles based on SD of simulated CPC intensity across years. Top quartile = counties most affected by the instrument. Birth/Abortion ratio = baseline birth rate / baseline abortion rate. $p$-values from two-sample $t$-test with unequal variances.

Figure A8: Extensive Margin: Share of Counties with Any CPC by Population




*Notes*: Share of counties with at least one CPC, by population sextile, 1990–2019.

Figure A9: Intensive Margin: CPC Intensity Conditional on Any CPC




*Notes*: Mean CPCs per 10,000 women among counties with at least one CPC, by population sextile, 1990–2019.

## E.1 Instrument Determinism and Stability

Table A9 decomposes the variation in the simulated instrument into components attributable to predetermined observables. The instrument is a function of $W_{c,t-1}$ by construction, so this exercise does not test the exclusion restriction (which follows from predetermination and the law of iterated expectations). Its purpose is to characterize where the instrument's variation comes from. County fixed effects alone explain 72 percent, confirming that most variation is cross-sectional. Adding year fixed effects raises this to 77 percent. The linear second-stage controls add 2 percentage points, and quadratic terms with interactions bring the total to 82 percent. The remaining 18 percent reflects simulation noise from finite $S = 1{,}000$ draws and higher-order nonlinear structure in the path-dependent logit that a second-order polynomial approximation cannot capture. Both components are either mechanically independent of structural errors (simulation noise) or still functions of $W_{c,t-1}$ (uncaptured nonlinearity).

Table A9: Instrument $R^2$ Decomposition

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Dependent variable: Simulated IV (CPCs per 10,000 women) | | | |
| $R^2$ | 0.7202 | 0.7728 | 0.7978 | 0.8154 |
| Adjusted $R^2$ | 0.7106 | 0.7634 | 0.7890 | 0.8068 |
| Observations | 4,380 | 4,380 | 4,380 | 4,380 |
| County FE | Yes | Yes | Yes | Yes |
| Year FE | No | Yes | Yes | Yes |
| $W_{c,t-1}$ (linear) | No | No | Yes | Yes |
| $W_{c,t-1}$ (quadratic + interactions) | No | No | No | Yes |

*Notes:* Each column regresses the simulated instrument on a progressively richer set of predetermined variables. Column (1) includes only county fixed effects. Column (4) adds all hazard model covariates with quadratic terms and pairwise interactions. The gap between the $R^2$ in column (4) and unity reflects Monte Carlo noise from finite simulation draws ($S = 1{,}000$) and nonlinear structure in the path-dependent logit not captured by a second-order polynomial approximation.

Table A10 provides an informal over-identification test. Under predetermination, the IV estimate should be stable when the instrument is constructed from different subsets of the conditioning set, because no single dimension of $W_{c,t-1}$ should be uniquely responsible for identification. The table re-estimates the baseline 2SLS specification after dropping each category of time-varying controls from the second stage. The estimates range from $-0.109$ (dropping economic controls) to $-0.177$ (dropping demographic controls), all significant at the 1 percent level with first-stage

$F > 160$. The variation across specifications is consistent with different control sets absorbing different amounts of omitted variable bias, but the sign and approximate magnitude are stable. If identification were driven by a single misspecified dimension of $W$—for example, if the instrument were picking up trends in religious composition that directly affect fertility—then dropping religious and political controls from the second stage should substantially alter the estimate. It does not.

Table A10: IV Stability Across Control Sets (Informal Over-Identification)

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Dep. var: Log abortion rate | | | | |
| CPCs per 10,000 women | $-0.144^{***}$ | $-0.109^{***}$ | $-0.177^{***}$ | $-0.157^{***}$ | $-0.137^{***}$ |
| | (0.040) | (0.037) | (0.040) | (0.042) | (0.041) |
| First-stage $F$ | 170.2 | 163.0 | 173.1 | 171.1 | 171.9 |
| Observations | 4,376 | 4,376 | 4,376 | 4,376 | 4,376 |
| Time-varying controls | Full | No econ. | No demog. | No polit./relig. | None |

*Notes:* Each column re-estimates the baseline 2SLS specification after dropping a category of time-varying controls from the second stage. The instrument is unchanged across columns. Column (1) is the baseline with the full control set. Column (2) drops unemployment. Column (3) drops non-white share and share aged 10–19. Column (4) drops Republican vote share, Protestant share, and Catholic share. Column (5) drops all time-varying controls (county and year FE only). All specifications use $\sqrt{\text{pop}}$ weights. Standard errors clustered at county level. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

# F Additional Results

## F.1 Age $\times$ Year Fixed Effects Specification

Table A11: Age-Specific CPC Effects: Pooled Specification with Age × Year FE

| | (1) AYFE Unweighted | (2) AYFE $\sqrt{\text{pop}}$ |
|---|---|---|
| *Panel A: Log Abortion Rate* | | |
| Ages 10-19 | -0.130*** | -0.170*** |
| | ( 0.047) | ( 0.050) |
| Ages 20-24 | -0.231*** | -0.261*** |
| | ( 0.043) | ( 0.045) |
| Ages 25-29 | -0.271*** | -0.288*** |
| | ( 0.042) | ( 0.047) |
| Ages 30-34 | -0.195*** | -0.246*** |
| | ( 0.044) | ( 0.044) |
| Ages 35-44 | -0.122*** | -0.167*** |
| | ( 0.030) | ( 0.037) |
| First-stage F | 23.2 | 22.9 |
| N | 17,000 | 17,000 |
| RMSE | 0.5370 | 0.4230 |
| *Panel B: Log Birth Rate* | | |
| Ages 10-19 | -0.020 | -0.011 |
| | ( 0.044) | ( 0.051) |
| Ages 20-24 | -0.009 | -0.006 |
| | ( 0.033) | ( 0.036) |
| Ages 25-29 | 0.047*** | 0.049** |
| | ( 0.017) | ( 0.023) |
| Ages 30-34 | 0.049** | 0.036 |
| | ( 0.021) | ( 0.030) |
| Ages 35-44 | 0.025 | 0.002 |
| | ( 0.025) | ( 0.030) |
| First-stage F | 23.2 | 22.9 |
| N | 17,000 | 17,000 |
| RMSE | 0.2895 | 0.2858 |
| *Panel C: Log Pregnancy Rate* | | |
| Ages 10-19 | -0.036 | -0.034 |
| | ( 0.041) | ( 0.046) |
| Ages 20-24 | -0.026 | -0.028 |
| | ( 0.031) | ( 0.033) |
| Ages 25-29 | 0.023 | 0.023 |
| | ( 0.016) | ( 0.022) |
| Ages 30-34 | 0.026 | 0.010 |
| | ( 0.020) | ( 0.027) |
| Ages 35-44 | 0.005 | -0.019 |
| | ( 0.019) | ( 0.023) |
| First-stage F | 23.2 | 22.9 |
| N | 17,000 | 17,000 |
| RMSE | 0.2524 | 0.2470 |

*Notes:* Each column pools all five age groups with age × year fixed effects (AYFE) and age-group-specific treatment interactions. Column (1) is unweighted; Column (2) uses $\sqrt{\text{population}}$ weights. The first-stage $F$ of ≈23 reflects dilution of treatment variation across pooled age groups (compared to $F > 110$ in the separate regressions, Table **??**). RMSE is the root mean squared error of the second stage. Standard errors clustered at county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## F.2 Revenue Heterogeneity

Table A12 splits the sample by median CPC funding (IRS Form 990 total revenue, CPI-adjusted). The first-stage $F$ of 8.0 in the above-median subsample (Columns 3–4) falls below the Stock and Yogo (2005) threshold of 16.4. Because the model is just-identified, weak instruments bias 2SLS toward OLS ($-0.072$), not away from zero, so the point estimate of $-0.497$ is if anything conservative. The Anderson-Rubin test, which is valid regardless of instrument strength, rejects the null of zero at the 5% level ($p = 0.031$). The Stock and Yogo (2005) critical values also assume i.i.d. errors and are conservative under clustering (Lee, McCrary, Moreira, & Porter, 2022).

Table A12: CPC Effects by Organizational Funding Level

| | (1) Log Abort | (2) Log Abort | (3) Log Abort | (4) Log Birth |
|---|---|---|---|---|
| CPCs per 10,000 women | -0.246** | -0.198* | -0.497*** | 0.202** |
| | (0.101) | (0.112) | (0.180) | (0.102) |
| Sample | Full (1996-2018) | Low funds + no CPC | High funds + no CPC | High funds + no CPC |
| N | 3,354 | 2,492 | 2,502 | 2,506 |
| First-stage F | 28.4 | 22.3 | 8.0 | 8.0 |

Standard errors in parentheses
County-year IV, log outcome, 1996–2018 (IRS filing years).
CPC funds from IRS Form 990, CPI-adjusted.
Median split based on total CPC funds among CPC county-years.
Col 2: only county-years with below-median or no CPC funding.
Col 3: only county-years with above-median CPC funding.
County and year FE, controls, $\sqrt{\text{population}}$ weights.
Standard errors clustered by county. First-stage $isKleibergen - Paap$.
$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

Table A13: CPC Effects with Organizational Funding Controls

| | (1) Log Abort | (2) Log Abort | (3) Log Abort |
|---|---|---|---|
| CPCs per 10,000 women | -0.104* | | -0.240** |
| | (0.060) | | (0.100) |
| CPC × Funds ($100k$) | -0.008 | | |
| | (0.014) | | |
| CPC funds ($100k$) | 0.007 | | 0.004 |
| | (0.006) | | (0.003) |
| CPC funds per 10k women | | -0.000 | |
| | | (0.000) | |
| Estimator | OLS | OLS | IV |
| N | 3,354 | 3,354 | 3,354 |
| Adj. R-sq | 0.5922 | 0.5886 | 0.0376 |
| First-stage F | | | 28.7 |

Standard errors in parentheses

County-year, 1996–2018 (IRS filing years).

CPC funds from IRS Form 990 (total revenue, CPI-adjusted).

Col 1: OLS with CPC×funding interaction.

Col 2: OLS with per-capita CPC funding as treatment.

Col 3: IV with funding as exogenous control.

County and year FE, controls, $\sqrt{\text{population}}$ weights.

Standard errors clustered by county.

$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

## F.3 Race Decomposition

Table A14: CPC Effects by Race and Age: Functional Form Comparison

| *Panel A: White* | | | | |
|---|---|---|---|---|
| | Poisson CF | | Log (2SLS) | |
| | Abortion | Birth | Abortion | Birth |
| Age 10–19 | -0.2147*** | -0.0798 | -0.1698** | -0.0912 |
| | (0.0637) | (0.0643) | (0.0742) | (0.0568) |
| Age 20–24 | -0.2583*** | -0.0415 | -0.2250*** | -0.0562* |
| | (0.0634) | (0.0352) | (0.0807) | (0.0302) |
| Age 25–29 | -0.1552*** | 0.0757*** | -0.1536*** | 0.0605*** |
| | (0.0423) | (0.0245) | (0.0578) | (0.0178) |
| Age 30–34 | -0.0886* | 0.0623** | -0.0522 | 0.0364 |
| | (0.0495) | (0.0246) | (0.0608) | (0.0226) |
| Age 35–44 | -0.1239** | 0.0320 | -0.0946** | 0.0439 |
| | (0.0505) | (0.0246) | (0.0448) | (0.0350) |
| *Panel B: Non-White* | | | | |
| | Poisson CF | | Log (2SLS) | |
| | Abortion | Birth | Abortion | Birth |
| Age 10–19 | -0.1695*** | 0.0145 | -0.2244*** | 0.0558 |
| | (0.0623) | (0.0474) | (0.0702) | (0.0540) |
| Age 20–24 | -0.1580*** | -0.0139 | -0.1319*** | 0.0401 |
| | (0.0544) | (0.0539) | (0.0456) | (0.0554) |
| Age 25–29 | -0.0492 | 0.0510 | -0.0933 | 0.0636 |
| | (0.0571) | (0.0626) | (0.0675) | (0.0598) |
| Age 30–34 | -0.0823 | 0.1572** | -0.1656** | 0.1349** |
| | (0.0630) | (0.0742) | (0.0767) | (0.0635) |
| Age 35–44 | -0.0715 | 0.1750 | -0.1255* | 0.0867 |
| | (0.0696) | (0.1139) | (0.0756) | (0.0945) |
| N per age group (White) | | 3,365 | | |
| N per age group (Non-White) | | 3,223 | | |

*Notes:* Each cell is from a separate regression on the race-specific subsample. Poisson CF: count outcome, exposure = race-age pop (semi-elasticities). Log: natural log of rate per 1,000 via 2SLS (zeros dropped). Treatment: CPCs per 10,000 county pop, instrumented by simulated CPCs. County and year FE, demographic controls, $\sqrt{\text{pop}}$ weights (2SLS). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

### F.4 Exploratory Evidence on Where Substituting Births Appear

Table A15 decomposes CPC effects by marital status and age group using the Poisson CF with marital-specific exposure (unmarried female population for unmarried outcomes, married female population for married outcomes).

Two patterns emerge. First, the abortion reductions are concentrated among unmarried women, with significant coefficients at ages 20–24 ($-0.320$, $p < 0.01$), 25–29 ($-0.216$, $p < 0.05$), and 35–44 ($-0.191$, $p < 0.05$). Unmarried teenagers show an abortion reduction ($-0.128$, $p < 0.05$) and a significant birth increase ($+0.184$, $p < 0.01$), producing the cleanest substitution cell in the data. Among unmarried adults, birth coefficients are near zero; the pregnancy null at the aggregate and age-specific levels rules out large-scale deterrence, and the near-zero unmarried birth coefficients instead reflect a detection problem in thin Poisson cells.

Second, the birth increases concentrate among married women. Married women aged 25–29 and 30–34 show significant birth increases ($+0.087$ and $+0.089$, both $p < 0.05$) without strong abortion reductions. One interpretation is that the marginal woman who carries to term instead of aborting is disproportionately likely to be in a partnership near desired fertility timing. An alternative is community spillovers. The data cannot distinguish these interpretations. Important caveats include the endogeneity of marital status and the interpolated denominators.

Table A15: Poisson CF: CPC Effects by Marital Status and Age Group

| | Unmarried | | Married | |
|---|---|---|---|---|
| | Abortion | Birth | Abortion | Birth |
| | (1) | (2) | (3) | (4) |
| *Panel A: Poisson CF (analytical SEs)* | | | | |
| Age 10–19 | -0.1283** | 0.1843*** | -0.4510** | -0.2073* |
| | (0.0574) | (0.0456) | (0.1874) | (0.1166) |
| Age 20–24 | -0.3203*** | 0.0047 | -0.0820 | 0.0934** |
| | (0.0851) | (0.0372) | (0.1071) | (0.0447) |
| Age 25–29 | -0.2161** | 0.0145 | -0.0886 | 0.0870** |
| | (0.1077) | (0.0364) | (0.0726) | (0.0404) |
| Age 30–34 | -0.1136 | 0.0343 | -0.0300 | 0.0885** |
| | (0.0767) | (0.0495) | (0.0554) | (0.0371) |
| Age 35–44 | -0.1908** | 0.0023 | -0.0498 | 0.0452 |
| | (0.0873) | (0.0507) | (0.0543) | (0.0391) |
| N per age group | 3,000 | | | |

*Notes:* Each cell is from a separate Poisson CF regression. Outcome is the marital-status-specific count; exposure is the corresponding marital-specific female population for the age group (unmarried exposure = unmarried female pop, married exposure = married female pop, from Census/ACS benchmarks). Coefficients are semi-elasticities of the marital-specific rate. Treatment is CPCs per 10,000 county population, instrumented via control function (Wooldridge 2014). Standard errors from ppmlhdfe, clustered by county. County and year FE, demographic controls. Sample restricted to county-years with observed marital status. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

### F.4.1 Marriage Balance and Power

Table A16: Balance Test: CPC Effects on Married Share of Women

| | OLS | 2SLS | SE | First-stage F | N |
|---|---|---|---|---|---|
| Pooled | -0.0058 | -0.0105 | ( 0.0070) | 133.5 | |
| Ages 10–19 | 0.0004 | 0.0036 | ( 0.0065) | 124.3 | 4380 |
| Ages 20–24 | -0.0293 | -0.0448*** | ( 0.0113) | 134.1 | 4380 |
| Ages 25–29 | 0.0007 | -0.0091 | ( 0.0105) | 132.4 | 4380 |
| Ages 30–34 | -0.0039 | -0.0014 | ( 0.0083) | 133.4 | 4380 |
| Ages 35–44 | 0.0031 | 0.0114** | ( 0.0047) | 132.9 | 4380 |
| Joint $\chi^2$ (5 df) | | 22.49 | | $p = 0.000$ | |

*Notes:* Dependent variable is the share of women who are married in each county-year-age cell. OLS and 2SLS specifications include county and year fixed effects, county demographic controls, and $\sqrt{\text{pop}}$ weights. Standard errors clustered by county. A zero or insignificant coefficient supports interpreting the marital decomposition (§6.6.2) as heterogeneity conditional on a predetermined characteristic.

Table A17: Ex-Post Power: Unmarried Birth Rate Coefficients

| | Estimate (1) | SE (2) | MDE (80%) (3) | Power at 11% (4) | Detectable? (5) |
|---|---|---|---|---|---|
| Age 10-19 | 0.1843 | (0.0456) | 0.1279 | 67.4% | No |
| Age 20-24 | 0.0047 | (0.0372) | 0.1043 | 84.1% | Yes |
| Age 25-29 | 0.0145 | (0.0364) | 0.1020 | 85.6% | Yes |
| Age 30-34 | 0.0343 | (0.0495) | 0.1387 | 60.3% | No |
| Age 35-44 | 0.0023 | (0.0507) | 0.1421 | 58.3% | No |

*Notes:* Column (1) reports the Poisson CF coefficient on CPCs per 10,000 women from the unmarried birth rate regression (Table A15, Panel A, Column 2). Column (3) is the minimum detectable effect at 80% power with 5% significance (two-sided): MDE $= (1.96 + 0.842) \times$ SE. Column (4) reports ex-post power to detect an 11% effect, the approximate magnitude implied by one-for-one substitution from unmarried abortions to unmarried births. "Yes" in Column (5) indicates the target effect exceeds the MDE.

### F.4.2 Substitution Ratios by Age Group

Table A18 reports birth-abortion substitution ratios by age group with delta-method confidence intervals. None reject the null of one-for-one substitution.

Table A18: Birth-Abortion Substitution Ratios by Age Group

| Age | $\hat{\beta}_A$ | $\hat{\beta}_B$ | $\bar{r}_A$ | $\bar{r}_B$ | Ratio | 95% CI | (= 1) |
|---|---|---|---|---|---|---|---|
| 10–19 | -0.169 | 0.017 | 7.2 | 23.9 | 0.34 | [-0.86, 1.54] | 0.281 |
| | ( 0.050) | ( 0.031) | | | ( 0.61) | | |
| 20–24 | -0.200 | -0.016 | 23.6 | 103.7 | 0.35 | [-0.62, 1.32] | 0.191 |
| | ( 0.043) | ( 0.022) | | | ( 0.49) | | |
| 25–29 | -0.212 | 0.036 | 16.2 | 88.9 | 0.93 | [-0.02, 1.88] | 0.881 |
| | ( 0.050) | ( 0.017) | | | ( 0.48) | | |
| 30–34 | -0.163 | 0.043 | 9.8 | 60.9 | 1.64 | [-0.00, 3.29] | 0.444 |
| | ( 0.044) | ( 0.019) | | | ( 0.84) | | |
| 35–44 | -0.136 | 0.052 | 3.5 | 14.8 | 1.64 | [ 0.02, 3.27] | 0.438 |
| | ( 0.039) | ( 0.022) | | | ( 0.83) | | |

*Notes:* Substitution ratio = $|\bar{r}_B \times \hat{\beta}_B| / |\bar{r}_A \times \hat{\beta}_A|$, where $\bar{r}$ is the population-weighted baseline rate per 1,000. SE and 95% CI via delta method (separate regressions, zero covariance). $-valuetests$: ratio = 1.0. All models: county and year FE, county controls, $\sqrt{\text{pop}}$ weights, clustered SEs. Outcomes are IHS-transformed aggregate rates per 1,000 (same specification as Table III).

# G Robustness Details

## G.1 Threats to Identification

**Anticipatory organizational responses.** If CPC organizations respond to anticipated future changes in fertility outcomes, then organizational timing would carry information about $u_{c,t}$, violating predetermination. The event study finds no evidence of differential pre-trends in abortion rates before CPC entry.

**Correlated changes in abortion provider access.** The second-stage specification controls for distance to the nearest abortion provider and provider counts. Results are stable when restricting to counties always more than 30 miles from the nearest provider ($-0.152$, $F = 149$; Table **??**).

**Persistent unobservables operating through lagged abortion rates.** County fixed effects absorb the time-invariant component. Placebo tests find no evidence of pre-trends. Including twice-lagged abortion rates as a second-stage control attenuates the IV coefficient from $-0.144$ to

$-0.098$ (Table **??**), a reduction consistent with some demand persistence but still economically meaningful.

**Functional form as the sole source of identifying variation.** The IV coefficient is insensitive to polynomial degree in the hazard model ($-0.138$ to $-0.145$; Table **??**).

## G.2 Trending Confounder Controls

Table A19: Robustness to County Characteristic × Year FE Interactions

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| CPCs per 10,000 women | -0.144*** | -0.136*** | -0.138*** | -0.138*** | -0.130*** | -0.133*** | -0.045 | -0.058 |
| | (0.040) | (0.038) | (0.039) | (0.038) | (0.036) | (0.036) | (0.035) | (0.039) |
| Interactions | None | Protestant × Year | Republican × Year | Unemp. × Year | All three × Year | All + P×R×Year | Col 6 + Nonwhite×Year | Col 7 + Log pop×Year |
| County FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| N | 4,376 | 4,376 | 4,376 | 4,376 | 4,376 | 4,376 | 4,376 | 4,376 |
| R-squared | 0.111 | 0.101 | 0.094 | 0.066 | 0.053 | 0.053 | 0.016 | 0.019 |
| First-stage F-stat | 170.176 | 159.934 | 169.482 | 172.467 | 163.606 | 161.582 | 186.271 | 183.448 |

*Notes:* Dependent variable is log abortion rate. All columns are IV using the simulated instrument. $X_c\times$ Year FE flexibly absorb county-specific trends along observable dimensions correlated with CPC expansion. Standard errors clustered at county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## G.3 Abortion Policy Landscape

Table A20: Robustness to State Abortion Policy Controls

| | Log Abortion Rate | | | | | Log Birth Rate | | |
|---|---|---|---|---|---|---|---|---|
| CPCs per 10,000 women | -0.144*** | -0.137*** | -0.138*** | -0.137*** | -0.068** | 0.019 | 0.010 | 0.010 |
| | (0.040) | (0.038) | (0.038) | (0.038) | (0.029) | (0.012) | (0.013) | (0.013) |
| Policy controls | No | TRAP | PI | All | All | No | TRAP | All |
| N | 4,376 | 4,376 | 4,376 | 4,376 | 4,376 | 4,380 | 4,380 | 4,380 |
| R-squared | 0.111 | 0.123 | 0.120 | 0.123 | 0.796 | 0.067 | 0.158 | 0.159 |
| First-stage F-stat | 170.176 | 168.954 | 169.007 | 168.859 | | 170.276 | 169.054 | 168.959 |

*Notes:* TRAP indicators are state × post-enactment dummies. PI indicators are state × post-enactment dummies for parental consent laws. Standard errors clustered at county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Border Spillover: Abortion and Birth Rates

*Panel A: Log Abortion Rate*

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| heightCPCs per 10,000 women | -0.144*** | -0.178*** | -0.166*** | -0.152*** | -0.143*** | -0.088** |
| | (0.040) | (0.056) | (0.048) | (0.045) | (0.040) | (0.042) |
| Sample | Full | Interior only | Drop VA border | Drop TN border | Drop GA border | Interior (TWFE) |
| N | 4,376 | 3,387 | 3,986 | 4,107 | 4,046 | 3,387 |
| R-squared | 0.111 | 0.122 | 0.104 | 0.126 | 0.112 | 0.807 |
| First-stage F-stat | 170.176 | 90.703 | 80.241 | 197.500 | 167.451 | |

*Panel B: Log Birth Rate*

| | (1) | (2) |
|---|---|---|
| heightCPCs per 10,000 women | 0.019 | -0.001 |
| | (0.012) | (0.017) |
| Sample | Full | Interior only |
| N | 4,380 | 3,390 |
| R-squared | 0.067 | 0.093 |
| First-stage F-stat | 170.276 | 90.833 |

*Notes:* All IV columns use the forward simulation instrument with county and year FE, controls, and $\sqrt{\text{pop}}$ weights. "Interior only" drops counties bordering VA, TN, or GA. Columns (3)–(5) drop counties bordering individual states. Column (6) reports TWFE for comparison. Standard errors clustered at county level. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

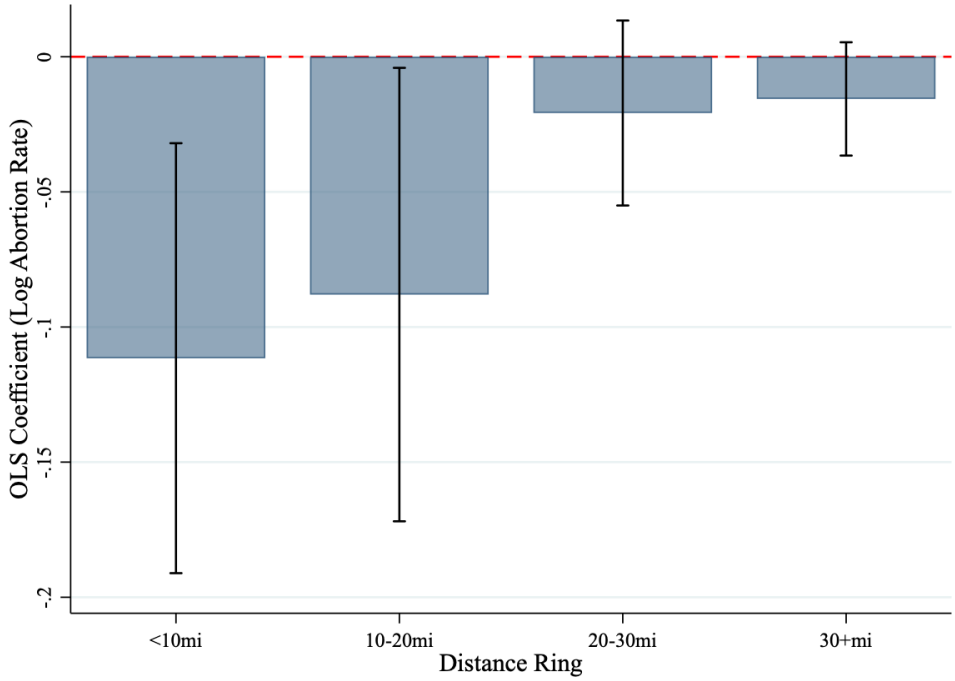



Figure A10: CPC Effect on Abortion Rate by Distance Ring (OLS)

*Notes*: OLS coefficients on CPC presence interacted with distance rings from the nearest CPC. Bars show 95% confidence intervals. County and year FE included.

## G.10 Oster Bounds on Selection

Table A26: Oster (2019) Bounds on Selection on Unobservables

| *Panel A: CPC Entry Hazard Model (Linear Probability Model)* | | | | | | |
|---|---|---|---|---|---|---|
| Variable | $\hat{\beta}_{short}$ | $\hat{\beta}_{long}$ | $R^2_{short}$ | $R^2_{long}$ | $\delta^*(R_{max}=1)$ | $\delta^*(1.3\tilde{R}^2)$ |
| Protestant share | 0.028 | 0.009 | 0.008 | 0.022 | 0.00 | 0.70 |
| Republican share | 0.007 | -0.013 | 0.007 | 0.022 | -0.01 | -1.28 |
| Log population | 0.017 | 0.018 | 0.015 | 0.022 | 0.02 | 1.59 |
| Dist. to clinic | -0.000 | 0.000 | 0.010 | 0.022 | -0.00 | -0.05 |
| Minority share | -0.048 | -0.048 | 0.010 | 0.022 | 0.01 | 1.07 |
| Abortion rate | 0.001 | 0.000 | 0.007 | 0.022 | 0.00 | 0.21 |
| *Panel B: OLS Structural Equation (Fertility Outcomes)* | | | | | | |
| Outcome | $\hat{\beta}_{short}$ | $\hat{\beta}_{long}$ | | $\tilde{R}^2$ | $\delta^*(R_{max}=1)$ | $\delta^*(1.3\tilde{R}^2)$ |
| Log abortion rate | -0.082 | -0.072 | | 0.119 | 0.68 | 13.34 |
| Log birth rate | 0.011 | 0.010 | | 0.068 | 0.91 | 38.45 |
| Log pregnancy rate | 0.011 | 0.010 | | 0.118 | 0.86 | 20.42 |
| *Panel C: Reduced Form (Instrument → Outcome)* | | | | | | |
| Outcome | $\hat{\pi}_{short}$ | $\hat{\pi}_{long}$ | | $\tilde{R}^2$ | $\delta^*(R_{max}=1)$ | $\delta^*(1.3\tilde{R}^2)$ |
| Log abortion rate | -0.113 | -0.121 | | 0.128 | 10.65 | 159.04 |
| Log birth rate | 0.022 | 0.016 | | 0.069 | 0.15 | 5.75 |
| Log pregnancy rate | 0.022 | 0.012 | | 0.118 | 0.14 | 3.30 |

*Notes:* $\delta^* > 1$ means unobservables would need to exceed observables to nullify the coefficient. Panel A uses a linear probability model without county FEs (Oster framework). Panel B residualizes county and year FEs before applying the Oster bounds procedure. Population-weighted. Panel C regresses outcomes directly on the simulated-CPC instrument (reduced form); large $\delta^*$ indicates unobservables would need to vastly exceed observables to explain the instrument–outcome relationship via a direct (non-CPC) channel, supporting the exclusion restriction.

## G.11 Plausibly Exogenous Instrument Bounds

Table A27: Plausibly Exogenous Instrument Bounds (2SLS)

| | UCI Confidence Set for $\beta$ | | |
|---|---|---|---|
| Direct effect support $[\underline{\delta}, \bar{\delta}]$ | Lower | Upper | Excludes zero? |
| *Panel A: Log Abortion Rate (Under 30)* | | | |
| Baseline IV (no direct effect) | -0.2482 ( | 0.0737) | Yes |
| $\delta \in [-0.02, 0.02]$ | -0.4152 | -0.0800 | Yes |
| $\delta \in [-0.05, 0.05]$ | -0.4576 | -0.0425 | Yes |
| $\delta \in [-0.10, 0.10]$ | -0.5246 | 0.0208 | No |
| *Panel B: Log Abortion Rate (Over 30)* | | | |
| Baseline IV (no direct effect) | -0.1969 ( | 0.0965) | Yes |
| $\delta \in [-0.02, 0.02]$ | -0.4076 | 0.0158 | No |
| $\delta \in [-0.05, 0.05]$ | -0.4488 | 0.0535 | No |
| $\delta \in [-0.10, 0.10]$ | -0.5137 | 0.1170 | No |

*Notes:* Union of Confidence Intervals (UCI) bounds following Conley et al. (2012). For each $\delta$ on a grid, the outcome is adjusted by $\tilde{y} = y - \delta z$ where $z$ is the simulated CPC density instrument, and 2SLS is re-estimated on the adjusted outcome. The UCI is the union of the resulting 95% confidence intervals across all $\delta$ values. "Excludes zero" indicates the entire UCI set does not contain zero. Both panels use log-transformed rates, $\sqrt{\text{pop}}$ weights, county and year fixed effects, and the standard control set. Under 30 pools ages 10–29; Over 30 pools ages 30–44.

## G.12 Instrument Leverage and Influence Diagnostics

Table A28: IV Estimates Excluding Predicted CPC Intensity Quartiles

| Sample | $\hat{\beta}_{\text{IV}}$ | SE | First-stage $F$ | $N$ | Counties |
|---|---|---|---|---|---|
| Full sample | -0.144 | (0.040) | 170.1 | 4,376 | 146 |
| Drop Q1 (lowest) | -0.142 | (0.043) | 163.7 | 3,269 | 109 |
| Drop Q4 (highest) | -0.129 | (0.109) | 69.4 | 3,297 | 110 |
| Drop Q1 & Q4 | -0.141 | (0.123) | 64.8 | 2,190 | 73 |

*Notes:* Counties ranked by mean predicted CPC intensity and assigned to quartiles. Each row re-estimates the baseline IV specification after excluding the indicated quartile(s).

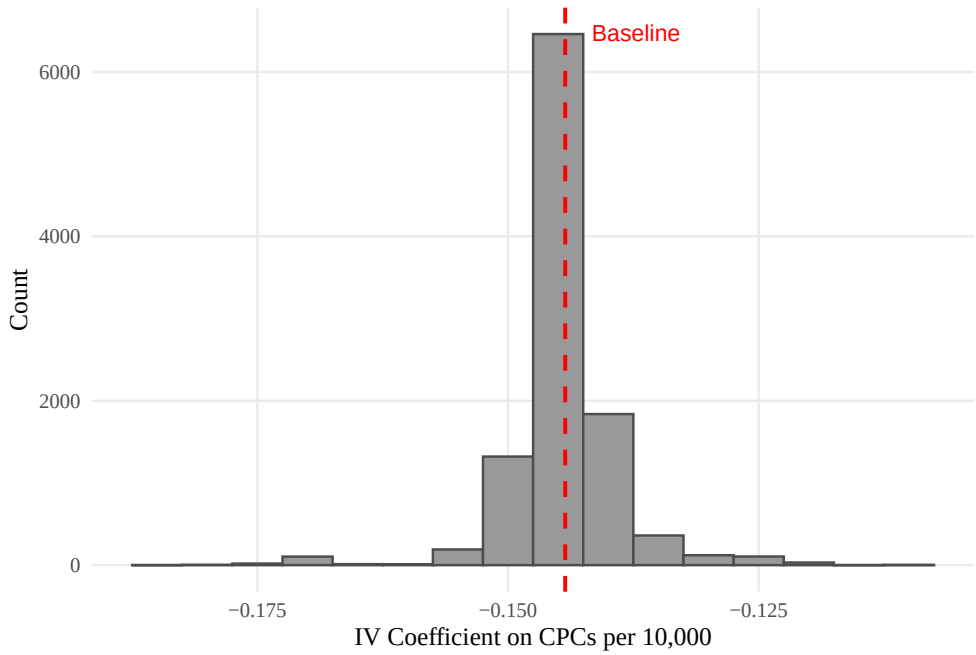
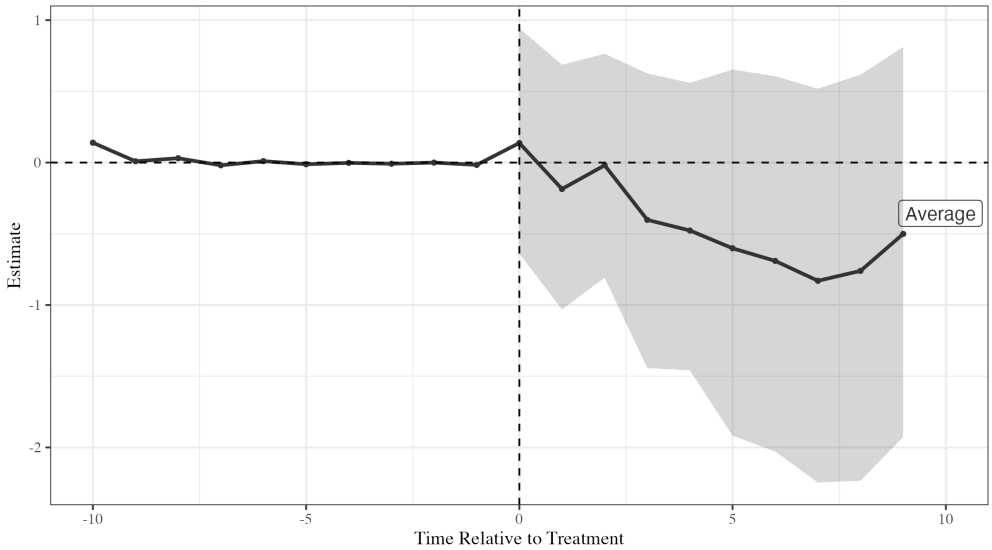



Figure A11: Leave-Two-County-Out IV Coefficient Distribution

*Notes:* Leave-two-county-out jackknife. No county pair shifts the coefficient by more than 20% from the baseline.

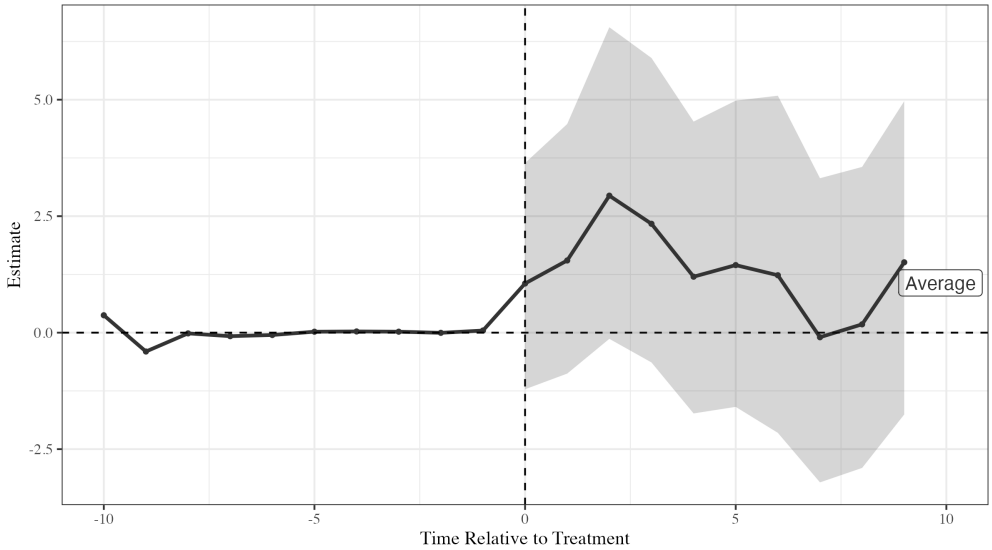
# H Synthetic Control

Figure A12: Augmented synthetic control: Abortion Rate (per 1,000 Women)



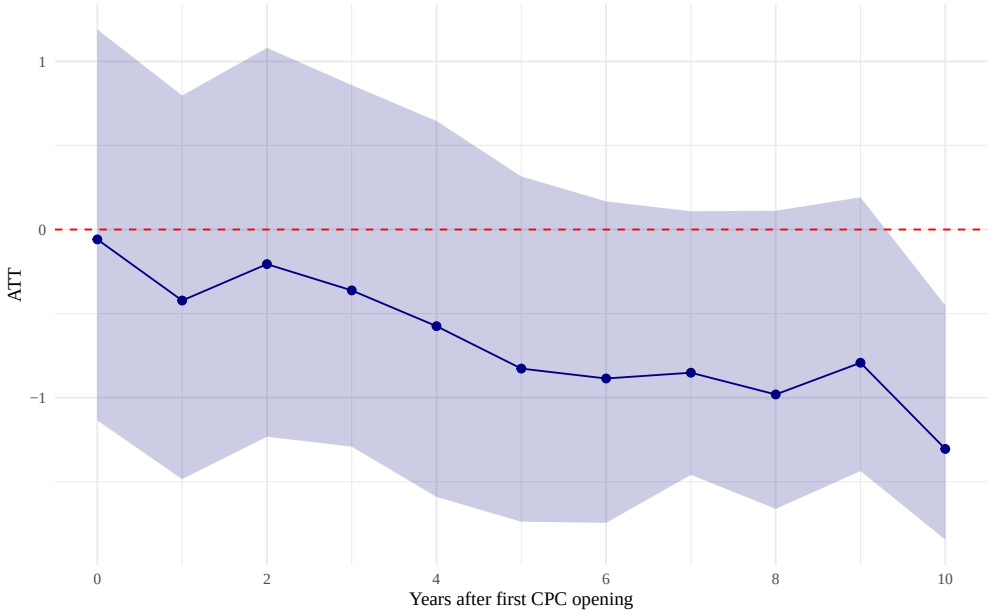

*Notes*: Augmented synthetic control estimates (Ben-Michael, Feller, & Rothstein, 2021) for the abortion rate in levels (per 1,000 women aged 10–44). Staggered adoption design with CPC opening as binary treatment. Levels are preferred over logs because the synthetic control convexity constraint is more naturally interpretable when applied to rates.

Figure A13: Augmented Synthetic Control: Birth Rate (per 1,000 Women)



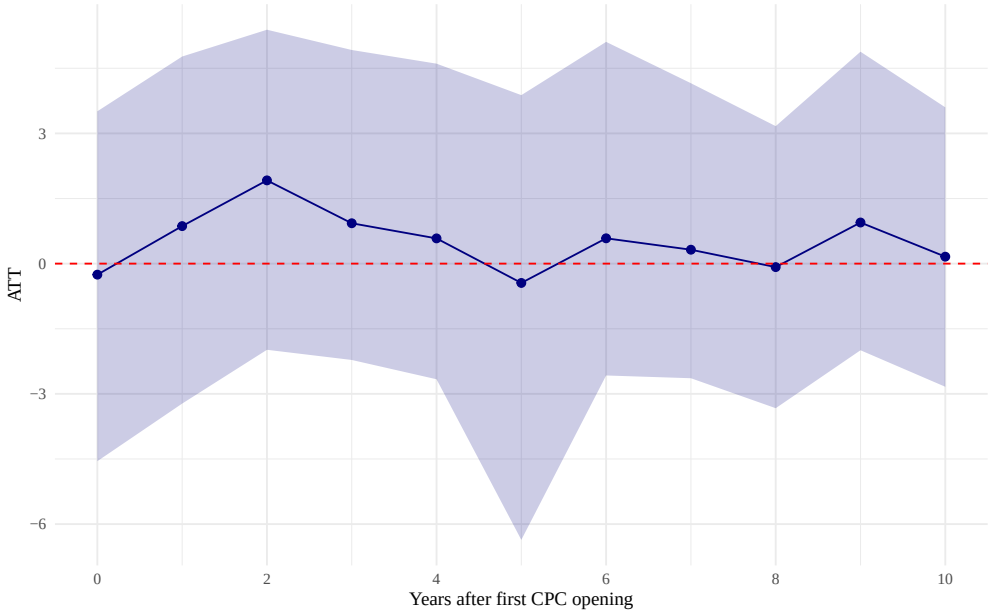

*Notes*: Augmented synthetic control estimates (Ben-Michael et al., 2021) for the birth rate in levels (per 1,000 women aged 10–44). Staggered adoption design with CPC opening as binary treatment.

Figure A14: SCPI Estimates: Abortion Rate (per 1,000 Women)




*Notes*: Synthetic control with prediction intervals (Cattaneo, Feng, & Titiunik, 2021). Outcome is the abortion rate in levels (per 1,000 women aged 10–44). 95% prediction intervals shown.

Figure A15: SCPI Estimates: Birth Rate (per 1,000 Women)




*Notes*: Synthetic control with prediction intervals (Cattaneo et al., 2021). Outcome is the birth rate in levels (per 1,000 women aged 10–44). 95% prediction intervals shown.

Table A29: Synthetic Control with Prediction Intervals (Cattaneo, Feng, and Titiunik 2021)

| | (1) Abort Rate | (2) Log Abort | (3) Birth Rate | (4) Log Birth |
|---|---|---|---|---|
| CPC (SC) | -0.6607 | -0.0692 | 0.5023 | 0.0046 |
| | [0.5018] | [0.0496] | [1.9104] | [0.0393] |
| Method | SC | SC | SC | SC |
| Inference | Pred. Int. | Pred. Int. | Pred. Int. | Pred. Int. |

*Notes:* Synthetic control estimates with prediction intervals following Cattaneo, Feng, and Titiunik (2021). Treatment is the presence of at least one CPC in the county (staggered adoption, absorbing). For each treatment cohort (counties first treated in the same year), treated counties are averaged into a single series and a synthetic control is constructed from the never-treated donor pool. Reported ATTs are cohort-size-weighted averages across treatment cohorts. Values in brackets are pseudo-standard errors derived from the width of the 95% prediction intervals (half-width / 1.96). SC weights are constrained to the simplex (non-negative, sum to one). Prediction intervals account for both in-sample uncertainty (pre-treatment fit) and out-of-sample uncertainty (extrapolation).

Table A30: Dose-Response Reconciliation: IV vs. Synthetic Control

| | Log Abortion | Log Birth |
|---|---|---|
| *Panel A: Method Comparison* | | |
| IV coefficient (per CPC/10k) | -0.144 | 0.019 |
| | ( 0.040) | ( 0.012) |
| Mean treatment intensity (treated cty-yrs) | 0.467 | |
| IV-implied binary ATT | -0.067 | 0.009 |
| SCPI binary ATT | -0.069 | 0.005 |
| | [ 0.049] | [ 0.040] |
| *Panel B: Level Decomposition (per CPC/10k)* | | |
| Baseline rate (per 1,000 women) | 11.07 | 45.87 |
| Implied change (per 1,000 women) | -1.60 | 0.87 |
| Share of prevented abortions | | 54.3% |

*Notes:* Panel A reconciles the IV per-unit dose-response estimates (Table II, Column 2) with the binary-treatment SCPI ATT. The IV estimates a marginal effect per CPC per 10,000 women. The SCPI compares counties with any CPC to never-treated counties, averaging over heterogeneous treatment intensities. The IV-implied binary ATT equals the IV coefficient multiplied by the mean treatment intensity among treated county-years (0.467 CPCs per 10,000, population-weighted). Panel B converts semi-elasticities to level changes using population-weighted baseline rates. The substitution share is the ratio of additional births to prevented abortions in level terms. Standard errors in parentheses (IV, clustered by county); pseudo-standard errors from prediction interval half-widths in brackets (SCPI).